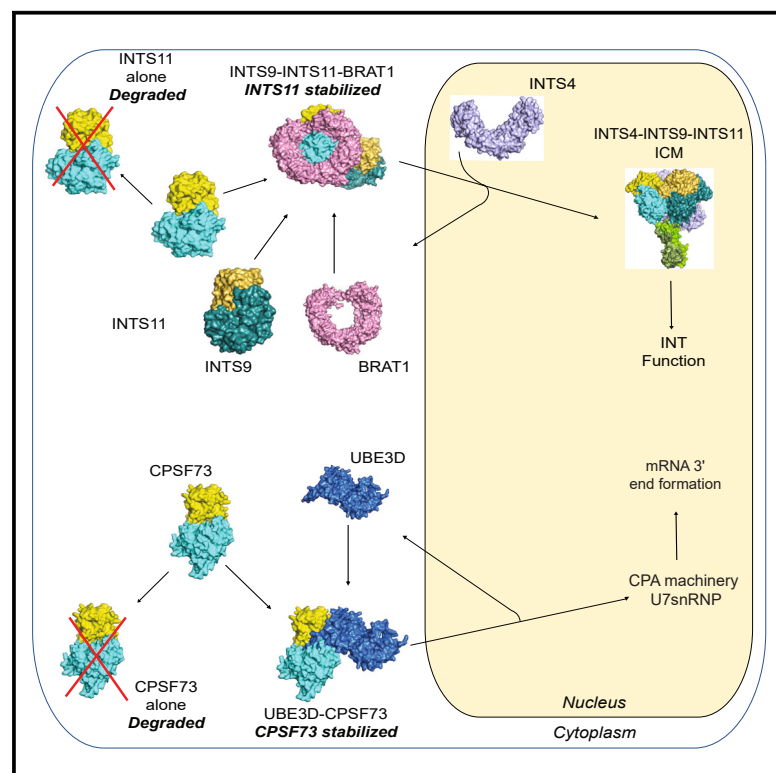


Cytoplasmic binding partners of the Integrator endonuclease INTS11 and its paralog CPSF73 are required for their nuclear function

Graphical abstract



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In brief

Lin et al. reveal BRAT1/CG7044 as a binding partner for the INTS11 endonuclease that behaves like a cytoplasmic chaperone and is required for Integrator function in the nucleus. UBE3D is also identified as a binding partner for the CPSF73 endonuclease in pre-mRNA 3'-end processing, with a similar mechanism of action.

Highlights

- BRAT1/CG7044 is a binding partner that stabilizes INTS11 in the cytoplasm
- BRAT1/CG7044 is required for Integrator function in the nucleus
- The conserved C terminus of BRAT1-CG7044 is captured in the INTS11 active site
- UBE3D is a binding partner of CPSF73 with a similar mechanism of action

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Article

Cytoplasmic binding partners of the Integrator endonuclease INTS11 and its paralog CPSF73 are required for their nuclear function

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SUMMARY

INTS11 and CPSF73 are metal-dependent endonucleases for Integrator and pre-mRNA 3'-end processing, respectively. Here, we show that the INTS11 binding partner BRAT1/CG7044, a factor important for neuronal fitness, stabilizes INTS11 in the cytoplasm and is required for Integrator function in the nucleus. Loss of BRAT1 in neural organoids leads to transcriptomic disruption and precocious expression of neurogenesis-driving transcription factors. The structures of the human INTS9-INTS11-BRAT1 and *Drosophila* dlntS11-CG7044 complexes reveal that the conserved C terminus of BRAT1/CG7044 is captured in the active site of INTS11, with a cysteine residue directly coordinating the metal ions. Inspired by these observations, we find that UBE3D is a binding partner for CPSF73, and UBE3D likely also uses a conserved cysteine residue to directly coordinate the active site metal ions. Our studies have revealed binding partners for INTS11 and CPSF73 that behave like cytoplasmic chaperones with a conserved impact on the nuclear functions of these enzymes.

INTRODUCTION

Transcription termination is an obligatory event in the RNA polymerase II (RNAPII) life cycle (reviewed in Eaton and West¹ and Rodríguez-Molina et al.²). Multiple mechanisms exist to ensure transcription termination occurs robustly at the proper location. Near the 3'-ends of protein-coding genes, "canonical termination" is carried out by the cleavage and polyadenylation (CPA) machinery that simultaneously causes the removal of RNAPII from the DNA template,² while the replication-dependent histone pre-mRNA 3'-end processing machinery (U7 small nuclear ribonucleoprotein particle [snRNP]) produces a non-polyadenylated mRNA.³ More recently, at the 5'-ends of protein-coding genes, it has been found that the Integrator complex (INT) is responsible for executing promoter-proximal termination,^{4–10} attenuating transcription. Despite significant differences in the constituency between INT and the CPA machinery, common mechanistic themes

are apparent. Both complexes incorporate phosphatase activities to reverse the effect of elongation stimulatory kinases (i.e., pTEFb)^{6,11–14} and utilize RNA endonucleases to cleave nascent RNA,^{4,7,9,15–19} which generates access points for exonucleases such as XRN2²⁰ to promote termination. Impairment of either enzymatic activities results in loss of efficient termination by INT¹⁰ or the CPA machinery.¹³

The role of INT in promoter-proximal termination has only recently been revealed.²¹ INT is a 17-subunit protein complex associated with RNAPII that is crucial for 3'-end processing of snRNAs²² and other noncoding RNAs,^{23–26} as well as premature termination through cleavage of nascent mRNA transcripts on paused, promoter-proximal RNAPII.^{4,5,21,27–29} Structural studies reveal that INT is composed of several modules,²¹ including a backbone that contains INTS1/2/7/12^{7,12,30} and a shoulder comprised of INTS5/8.^{31,32} These two modules form a scaffold upon which two enzymatic factors associate: the Integrator phosphatase module containing PP2A/C/INTS6^{6,11} and the

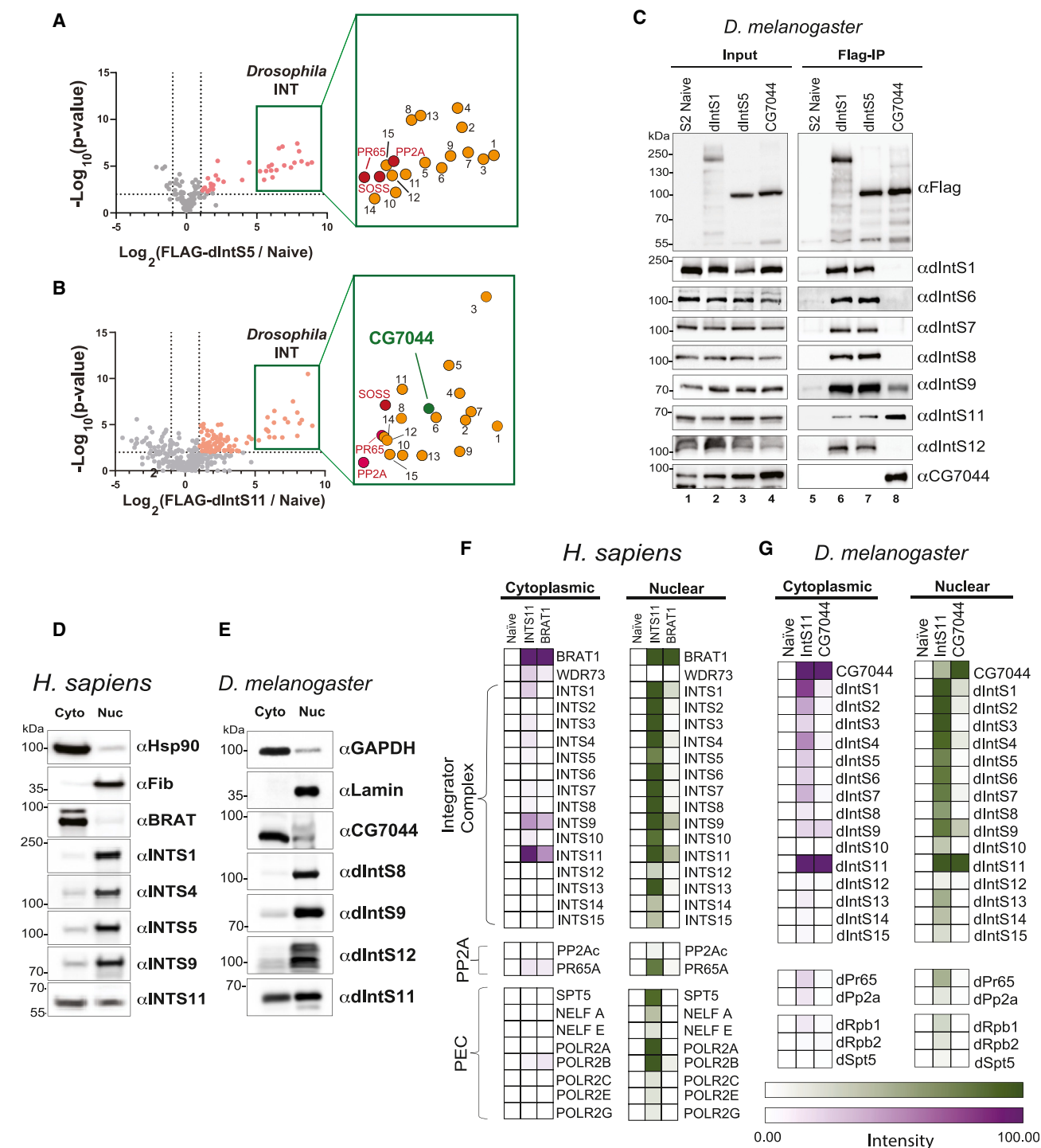


Figure 1. BRAT1/CG7044 is a conserved, cytoplasmic factor associated with INTS11

(A) Volcano plot of proteomics data analyzed from purifications using FLAG-dIntS5 relative to naive control. Integrator subunits are labeled in orange in the expanded view, and Integrator-associated proteins in red. Proteins in pink are statistically significant, and those in gray are not significant or unenriched.

(B) Volcano plot of proteomics data analyzed from purifications using FLAG-dIntS11 relative to naive control.

(C) Western blot analysis of input cellular extracts (left) and IP (right) from *Drosophila* S2 cells stably expressing FLAG-tagged proteins as indicated. IP conducted using anti-FLAG affinity resin was normalized to FLAG signals in each IP.

(D) Western blot of indicated endogenous proteins from cellular fractionation of HEK293T cells. The nuclear fraction is identified with fibrillarin (Fib) and cytoplasmic fraction with heat shock protein 90 (Hsp90).

(legend continued on next page)

Integrator cleavage module (ICM) containing INTS4/9/11.^{31,33,34} INTS9 and INTS11 are paralogs of CPSF100 and CPSF73 in the CPA machinery and the U7 snRNP,³⁵ with CPSF73 catalyzing the cleavage reaction in these machineries,^{16,36,37} and INTS11 being the endonuclease for Integrator.²² CPSF73 and INTS11 belong to the metallo- β -lactamase-associated CPSF, Artemis, SNM1/PSO2 (β -CASP) family of nucleases,³⁸ and their active sites are between a metallo- β -lactamase and a β -CASP domain, with two metal ions critical for catalysis. CPSF73 and INTS11 exist in a closed, inactive conformation on their own. They transition to an open conformation for catalysis, with a deep canyon between the metallo- β -lactamase and β -CASP domains to bind the RNA substrate.

Integrator has also been reported to interact with other factors, including a collection of zinc finger proteins,^{39,40} BRAT1,⁴¹ WDR73,⁴² and the SOSS complex.^{43–45} BRAT1 (CG7044 in *Drosophila*) was initially identified by its association with BRCA1 and ataxia telangiectasia mutated (ATM),⁴⁶ and mutations of BRAT1 have been linked to several neurological diseases.^{47,48} Loss of BRAT1 disrupts UsnRNA 3'-end processing and premature termination at nascent mRNAs, which indicates its expression is somehow required for INT function.⁴¹ The neurological disruptions observed in BRAT1 patients have parallels with symptoms associated with variants of WDR73 and INT subunits,^{42,49–51} suggesting a common requirement during neuronal differentiation.

We show here that BRAT1 and CG7044 are primarily cytoplasmic proteins distinct from the nuclear INT. Using cryo-electron microscopy (cryo-EM), we determined the structures of *Drosophila* dlntS11-CG7044 and human INTS9-INTS11-BRAT1 complexes. The structures reveal, unexpectedly, that the conserved C terminus of BRAT1/CG7044 is captured in the active site of INTS11, with INTS11 in a previously unobserved, "semi-open" state to accommodate this C terminus. Moreover, the conserved penultimate Cys residue is directly coordinated to the metal ions in the active site. BRAT1 engagement to the INTS11 active site is critical to INT function as mutations of the BRAT1 C terminus prevent rescue of INT loss of function in BRAT1 null cell lines. Further, loss of BRAT1 in neural organoids (NOs) leads to transcriptomic disruption and precocious expression of neurogenesis-driving transcription factors. Our observations with INTS11 inspired us to search for a similar partner for CPSF73. We found that UBE3D can form a stable complex with CPSF73, and a long segment on the surface of UBE3D likely extends into the CPSF73 active site, remarkably also with a conserved Cys residue likely directly coordinating the metal ions. These studies demonstrate the unexpected existence of cytoplasmic partners for INTS11 and CPSF73 that may share conserved features in their binding to the cognate endonucleases.

RESULTS

BRAT1/CG7044 is a conserved, primarily cytoplasmic factor associated with INTS11

Structural and biochemical analyses of Integrator have revealed its modular nature,^{12,15,31,33,34,52} but little is known about the biochemical, structural, and physiological relevance of Integrator-associated factors such as BRAT1.⁴¹ We generated total cell extracts from S2 cell lines stably expressing FLAG-tagged dlntS5, dlntS11, or empty vector as a control. As expected, using affinity purification followed by liquid chromatography-mass spectrometry (LC-MS), we observed strong enrichment of all 17 *Drosophila* Integrator subunits in both FLAG-dlntS5 (Figure 1A) and FLAG-dlntS11 relative to control (Figure 1B). Interestingly, CG7044 was only observed in FLAG-dlntS11 purification (Figure 1B; Table S1). We extended this analysis to complexes associated with FLAG-dlntS1 or FLAG-dlntS8 and used LC-MS (Figure S1A; Table S1) and western blotting (Figure S1B) to confirm that CG7044 is only associated with dlntS11. Finally, we generated a reciprocal S2 cell line stably expressing FLAG-CG7044 and compared associated complexes to those observed from the purification of FLAG-dlntS1, FLAG-dlntS5, or control. Using western blot (Figure 1C) and LC-MS analysis (Figure S1C; Table S1), we found that CG7044 associates strongly with dlntS11 and, to a lesser extent, dlntS9, but no other Integrator subunits were observed.

BRAT1 was initially described as a mostly cytoplasmic protein capable of shuttling into the nucleus,⁵³ but this has yet to be assessed in the context of Integrator localization and function, which is thought to be nuclear. To address this, we first fractionated human HEK293T cells into nuclear and cytoplasmic extracts (Figure 1D). As expected, most Integrator subunits display predominantly nuclear localization except INTS11, which is observed in both compartments as consistent with previous reports.⁵⁴ By contrast, endogenous BRAT1 is primarily cytoplasmic, suggesting its function may be distinct from that of the nuclear Integrator. The prevalent cytoplasmic localization of BRAT1 was confirmed using indirect immunofluorescence (Figure S1D). Our observations on INTS11 and BRAT1 in human cells are conserved in *Drosophila* cells (Figure 1E).

Given the cellular localization of BRAT1/CG7044 and INTS11 within human and fly cells, we used proteomics analysis to identify associated proteins in either compartment. To that end, we affinity purified complexes from cytoplasmic or nuclear extracts from HEK293T and S2 cells stably expressing exogenous FLAG-tagged INTS11, BRAT1/CG7044, or control. In the cytoplasm, human FLAG-INTS11 and FLAG-BRAT1 interact with each other, INTS9, and another Integrator-associated protein, WDR73 (Figure 1F; Table S1). In *Drosophila*, we found that FLAG-dlntS11 and FLAG-CG7044 also interact with each other

(E) Similar to (D), except *Drosophila* S2 cells were used to show conservation of localization. The nuclear fraction is identified with lamin and cytoplasmic fraction with GAPDH.

(F) Heatmap derived from immunoprecipitation (IP) LC-MS analysis of FLAG IP from HEK293T cells expressing the indicated FLAG-tagged proteins. Intensity bars reflect mass spectrometry (MS) intensity normalized to the immunoprecipitated FLAG protein in each column and then quantified as relative to control. The actual BRAT1 level in the nucleus (right column) is approximately 5-fold lower than INTS11 level (middle column). They have similar colors in the plot due to the normalization. Green: nuclear; purple: cytoplasmic. Control cells lack any exogenously expressed FLAG-tagged protein.

(G) Heatmap derived from IP LC-MS analysis of FLAG IP of *Drosophila* S2 cells expressing indicated FLAG-tagged proteins.

in the cytoplasm and a smaller amount of dIntS9 (Figure 1G; Table S1). Notably, no other protein was significantly enriched in the fly dIntS11-CG7044 complex, and no readily identifiable ortholog exists for WDR73, suggesting it may be a vertebrate-specific interaction. In the nucleus, both human and fly FLAG-INTS11 are observed to interact with the entire INT, several members of the pre-elongation complex, and BRAT1. On the contrary, FLAG-tagged BRAT1/CG7044 in the nucleus was still only associated with INTS9 and INTS11 (Figures 1F and 1G). Altogether, these results indicate that a distinct “pool” of INTS9 and INTS11 is associated with BRAT1/CG7044 primarily in the cytoplasm, raising the intriguing question of how this cytoplasmic complex contributes to the functions of Integrator in the nucleus.

Structure of the dIntS11-CG7044 complex

To gain molecular insight into the *Drosophila* dIntS11-CG7044 association, we co-expressed and purified their complex (Figure 2A). We also co-expressed dIntS9-dIntS11-CG7044 (Figures 1E and 1G) but found that dIntS9 was present at sub-stoichiometric levels after purification even though its expression level was very high (Figure 2B), consistent with the immunoprecipitation data.

We determined the structure of the dIntS11-CG7044 complex at 3.54 Å resolution by cryo-EM (Figures S2A–S2D; Table 1). Most CG7044 and dIntS11 residues could be located with good side-chain density (Figure 2C). The structure reveals that CG7044 primarily forms a large Arm/HEAT domain (Figures 2D–2G), with the pairs of anti-parallel helices arranged in the shape of a horseshoe that wraps around dIntS11 (Figures 2D and 2E). The connection between the two helices in the last repeat is exceptionally long, containing residues 830–920 and forming a “lasso” structure (Figure 2C). The C-terminal extension (CTE) of CG7044 beyond the Arm/HEAT domain traverses the open end of the horseshoe, contacting the N-terminal region of CG7044 (Figures 2D and 2F). The dIntS11-CG7044 interface is extensive, with $\sim 3,700$ Å² buried surface area for each protein, consistent with the stability of this complex. The Arm/HEAT domain of CG7044 contributes 1,750 Å² to the buried surface area.

The conserved CG7044 DCY motif is captured in the dIntS11 active site

The structure unexpectedly reveals that the C terminus of CG7044 is inserted into the active site region of dIntS11 (Figure 3A). The last three residues of CG7044, Asp-Cys-Tyr974 (DCY), are conserved among all known CG7044/BRAT1 homologs (Figure 3B), further emphasizing their importance. This DCY motif resides in a deep pocket at the interface between the dIntS11 metallo- β -lactamase and β -CASP domains. The side chain of Asp972 is hydrogen-bonded to His392 of dIntS11 (Figure 3C), the general acid for the nuclease reaction.^{16,37} The side chain of Cys973 is directly coordinated to both metal ions in the active site, which would not allow the coordination by the scissile phosphate of the RNA substrate.³⁷ Finally, the carboxylate group at the C terminus of CG7044 has ion-pair interactions with Arg244 in the β -CASP domain. There is no space to accommodate additional amino acid residues following Tyr974 (Figures 3A and 3C).

Based on these observations, the C-terminal residues of CG7044 would directly compete against the RNA substrate. In support of this, the binding mode of DCY extensively overlaps with that of the RNA substrate observed in the structure of CPSF73 (Figure 3D).³⁷ Overall, the structural observations demonstrate that CG7044 forms a tight complex with dIntS11 and can inhibit its nuclease activity.

dIntS11 is in a semi-open state in complex with CG7044

We and others have previously established that INTS11 in the ICM alone is in an inactive, closed state.^{12,31,34} When we compared the structure of dIntS11 in the CG7044 complex with that in the *Drosophila* ICM,³⁴ we found that there is a substantial conformational difference for the β -CASP domain, corresponding to a rotation of 6.3° relative to the metallo- β -lactamase domain (Figure 3E). This change is distinct from that for the open-closed transitions of CPSF73³⁷ and human INTS11,^{7,15,34} which has a 17° rotation for the β -CASP domain (Figure S3A), and does not create a canyon for RNA binding. In the ICM, the binding pocket for the C terminus of CG7044 does not exist due to the closed state of IntS11 (Figures S3B and S3C). Therefore, dIntS11 assumes a new state in the CG7044 complex, and we will refer to it as a semi-open state. In this state, there is no canyon between the metallo- β -lactamase and β -CASP domains that would allow the RNA substrate to reach the active site (Figures S3D and S3E).

The CG7044 lasso interacts with the β -CASP domain of dIntS11

The extended connection between the two helices in the last helical repeat of CG7044 forms a lasso structure, with two helices at the tip of this lasso being projected more than 30 Å away from the body of CG7044 (Figures 2E and 2G). The CG7044 lasso has extensive, predominantly hydrophobic interactions with the β -CASP domain of dIntS11. Residues in the two helices at the tip of the lasso (α L1 and α L2) contact the “back” face of the β -CASP domain (Figures 2E, S4A, and S4B). Most residues in the two helices interacting with dIntS11 are highly conserved among insect CG7044 homologs (Figure S4C). On the other hand, the linkers from these two helices to the Arm/HEAT domain are poorly conserved, suggesting that this structural feature is likely highly dynamic.

There is a significant conformational change for helices α C and α D of the dIntS11 β -CASP domain in the CG7044 complex (Figure S4D). Compared with the *Drosophila* ICM structure, the α D helix is disordered in the CG7044 complex, and the α L2 helix of the lasso takes up the position of this helix. The α L1 helix of the lasso clashes with dIntS4 of the ICM, and a segment in the β -CASP domain of dIntS9 in the ICM clashes with CG7044. Therefore, the dIntS11-CG7044 complex is incompatible with the formation of the ICM, indicating that dIntS11 binding to CG7044 is mutually exclusive to its binding to dIntS4/dIntS9.

Structure of the human INTS9-INTS11-BRAT1 complex

Our biochemical studies with the *Drosophila* dIntS11-CG7044 complex suggest that a human INTS11-BRAT1 complex is also likely to exist. While we were able to purify this complex, the expression levels of the two proteins were very low, even though they were co-expressed (Figure 4A). However, when we included

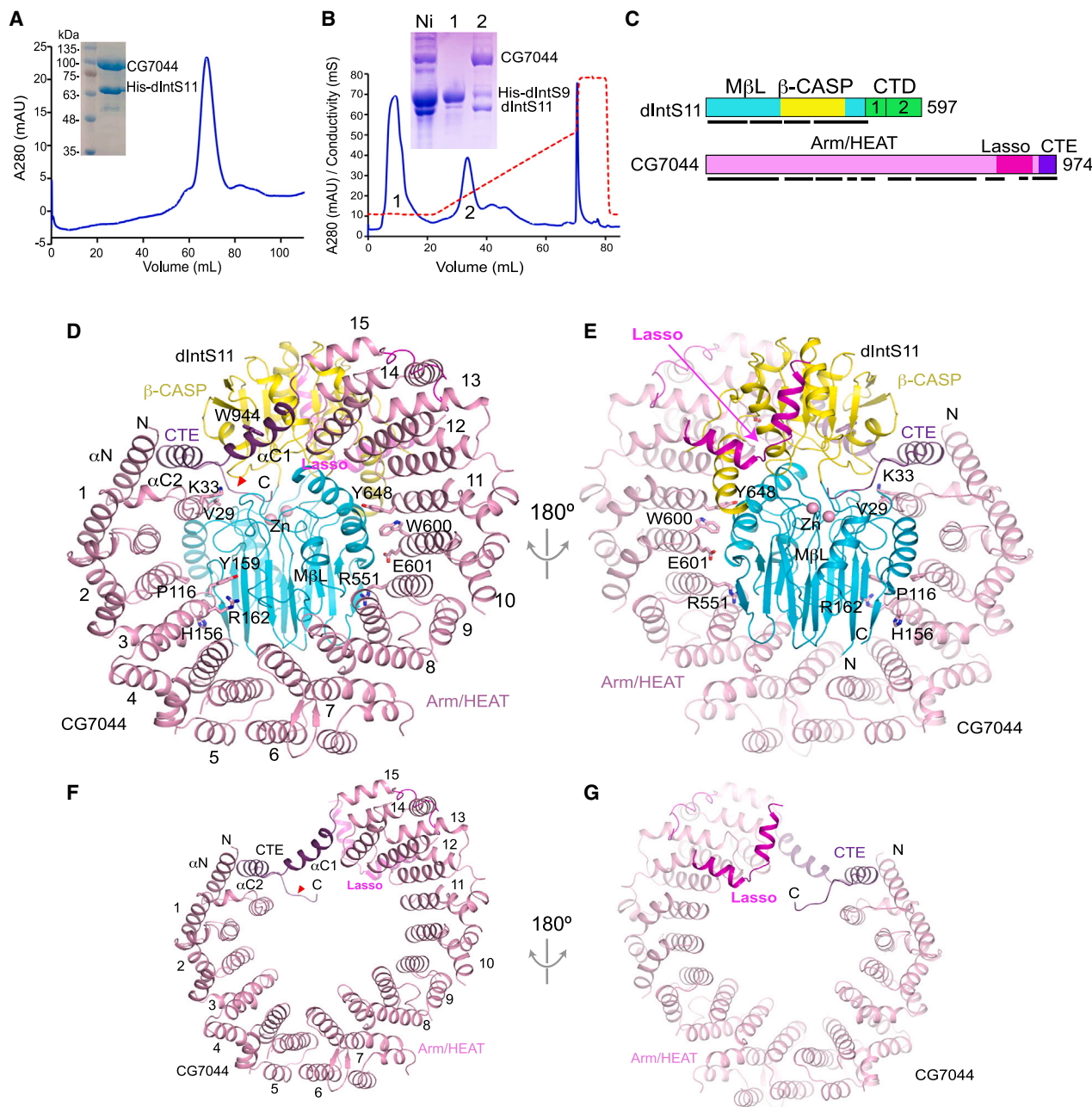


Figure 2. The overall structure of the *Drosophila* dlntS11-CG7044 complex

(A) Gel filtration profile of *Drosophila* dlntS11-CG7044 complex. Inset: SDS-PAGE gel of the complex.

(B) Ion exchange chromatogram of *Drosophila* dlntS9-dlntS11-CG7044 complex. Inset: SDS-PAGE gel of fractions from peaks 1 and 2 in the chromatogram and eluate from the nickel affinity column (Ni).

(C) Domain organizations of *Drosophila* dlntS11 and CG7044. Residues observed in the structure are indicated with black lines. MβL, metallo-β-lactamase; β-CASP, metallo-β-lactamase-associated CPSF, Artemis, SNM1/PSO2; CTD, C-terminal domain; CTE, C-terminal extension.

(D) The overall structure of the dlntS11-CG7044 complex. Domains are colored as in (C). The red arrowhead indicates the C terminus of CG7044 in the active site of dlntS11. CG7044 residues with >50 Å² buried surface area in the interface with dlntS11 are shown in stick models and labeled.

(E) The overall structure after 180° rotation around the vertical axis. The back face of dlntS11 and the lasso of CG7044 are visible.

(F and G) The structure of CG7044 alone, in the same view as (D) and (E). The structure figures were produced with PyMOL (www.pymol.org) unless indicated otherwise.

Table 1. Cryo-EM data collection, structure refinement, and validation statistics

	<i>Drosophila</i> dIntS11- CG7044 complex (PDB: 8UIC)	Human INTS9- INTS11- BRAT1 complex (PDB: 8UIB)
Data collection and processing		
Magnification	81,000	105,000
Voltage (kV)	300	300
Electron exposure (e ⁻ /Å ²)	51	58.2
Defocus range (μm)	−1 to −2.5	−0.8 to −1.3
Pixel size (Å)	1.083	0.83
Symmetry imposed	C1	C1
Image stacks (no.)	1,603	4,030
Initial particle images (no.)	2,028,174	1,112,411
Final particle images (no.)	406,222	185,172
Map resolution (Å)	3.54	3.21
FSC threshold	0.143	0.143
Map sharpening B-factor (Å ²)	−171.7	−126.1
Refinement		
Number of protein residues	1,227	1,534
Number of metal ions	2	2
Number of atoms	9,968	11,876
RMSDs		
Bond lengths (Å)	0.006	0.006
Bond angles (°)	0.865	0.863
PDB validation		
Clash score	6.78	9.84
Poor rotamers (%)	0.00	0.00
Ramachandran plot		
Favored (%)	91.0	93.0
Allowed (%)	9.0	7.0
Disallowed (%)	0.0	0.0

INTS9 in the co-expression, it led to a significant improvement in the expression levels of the INTS9-INTS11-BRAT1 complex (Figure 4B), consistent with pull-down assays (Figure 1F). We determined the structure of this ternary INTS9-INTS11-BRAT1 complex at 3.2 Å resolution by cryo-EM (Figures 4C–4E and S2E–S2H; Table 1).

The overall structure of the human INTS11-BRAT1 portion of the ternary complex is similar to that of the *Drosophila* dIntS11-CG7044 complex (Figure S5A). Human INTS11 is also in a semi-open state, and the BRAT1 C terminus is captured in the active site region of INTS11 with a similar binding mode (Figure 4F). With INTS11 in superposition, a rotation of 4° is needed to align the structure of BRAT1 with that of CG7044 (Figure S5A), with substantial differences for some of the Arm/HEAT repeats (Figure S5B). The lasso is not present in human BRAT1, as that

loop is much shorter (residues 733–772) and is disordered (Figure 4C). Therefore, the lasso may be a feature unique to CG7044 homologs in insects. Nonetheless, helix αD of the INTS11 β-CASP domain is also disordered in this complex (Figure 4E), suggesting that the disordering of this helix is a consequence of the binding of BRAT1/CG7044 rather than the presence of the lasso.

INTS9 has weaker interactions with BRAT1

INTS9 is bound to the side of BRAT1 and has essentially no direct contact with INTS11 (buried surface area of <60 Å²) in this complex (Figures 4D, 4E, and S5C). The β-CASP domain of INTS9 is located in the interface with BRAT1. By contrast, much of its metallo-β-lactamase domain is not in the interface (Figure S5C). The INTS9-BRAT1 interface is much less extensive than the INTS11-BRAT1 interface, suggesting that the INTS9-BRAT1 complex is less stable. A large, primarily hydrophobic interface patch involves helices αA and αB of the INTS9 β-CASP domain contacting repeats 10 and 11 of BRAT1 (Figure S5D).

Compared with the structure of INTS9 in the human ICM,³¹ a large conformational change is observed for helix αA and the neighboring strand βI of the β-CASP domain (Figure S5E). More importantly, the organization of INTS9 and INTS11 in the complex with BRAT1 is incompatible with the observed organization of the ICM. This can be visualized by overlaying the metallo-β-lactamase domains of INTS11 in the ICM and the INTS9-INTS11-BRAT1 complex (Figure 4G; Video S1). In this overlay, there are substantial differences in the position and orientation of INTS9 between the two complexes. INTS9 in the INTS9-INTS11-BRAT1 complex has essentially no contacts with INTS11, and it clashes with INTS4 in the ICM. This is supported by our observation that BRAT1 only pulls down INTS11 and INTS9 in both the nucleus and the cytoplasm (Figure 1F).

We investigated why *Drosophila* dIntS9 has a weaker association with the dIntS11-CG7044 complex (Figure 2B). We looked at how *Drosophila* dIntS9 and CG7044 would fit in the observed human INTS9-BRAT1 interface. Residues in this interface are generally weakly conserved among their homologs. In particular, Ser283 in human INTS9 is replaced by Met282 in *Drosophila* dIntS9 (Figure S5F), which would clash with CG7044 and destabilize the dIntS9-CG7044 complex. Based on these structural analyses, we generated the S283M single mutant as well as the E280R/S283M/L287A/R290A quadruple mutant of human INTS9 and found they were not able to disrupt the formation of the ternary complex based on gel filtration (Figure S5G). However, the C-terminal domains (CTDs) of INTS9 and INTS11 are likely still associated in the mutants,^{31,34} which would give an apparent ternary complex on gel filtration. We then examined the E280R/S283M/L287A/R290A sample by cryo-EM and found that the particles look like dIntS11-CG7044 hetero-dimers rather than INTS9-INTS11-BRAT1 hetero-trimers (Figure S5H), demonstrating that the mutation has disrupted the INTS9-BRAT1 interaction.

Functional relevance of CG7044/BRAT1-containing complexes

To examine the importance of BRAT1/CG7044 in human and *Drosophila* cells, we reduced its expression using either CRISPR-Cas9 or RNA interference. To that end, we generated

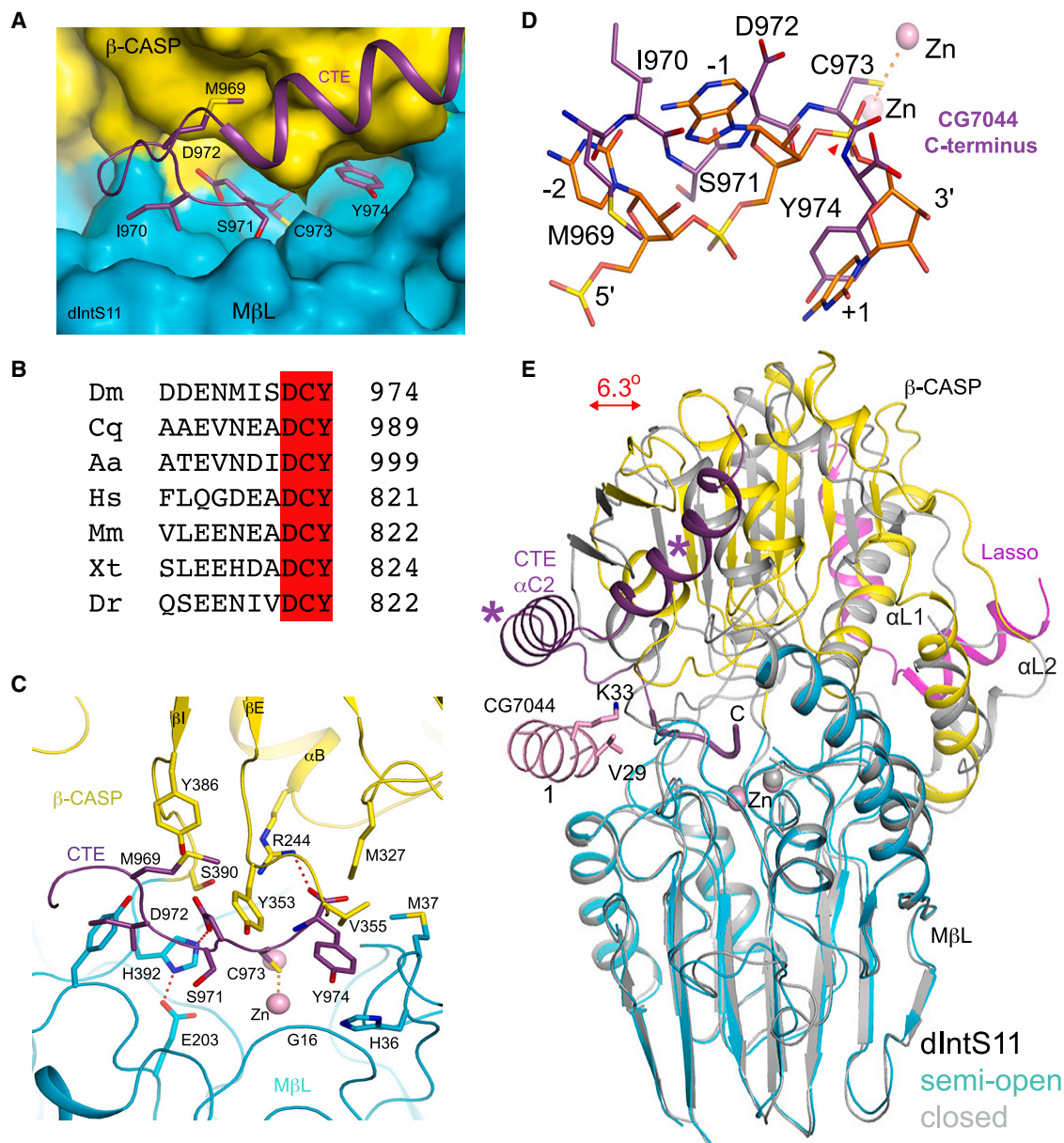


Figure 3. The conserved C terminus of CG7044 is captured in the active site of dIntS11

(A) Residues 972–974 at the C terminus of CG7044 (violet) are located in a pocket at the interface between the metallo-β-lactamase and β-CASP domains of dIntS11.

(B) Alignment of the C-terminal sequences of selected CG7044 and BRAT1 homologs. Conserved residues are highlighted in red. Dm, *D. melanogaster*; Cq, *Culex quinquefasciatus*; Aa, *Aedes aegypti*; Hs, *Homo sapiens* (human); Mm, *Mus musculus* (mouse); Xt, *Xenopus tropicalis* (frog); and Dr, *Danio rerio* (zebrafish).

(C) Detailed interactions between the C terminus of CG7044 and dIntS11 active site. Hydrogen-bonding interactions are indicated with the dashed lines in red.

(D) Overlay of the binding mode of the C terminus of CG7044 (violet) with that of the histone pre-mRNA substrate in human CPSF73 (orange).

(E) Overlay of the semi-open state of dIntS11 (in color) and the closed state of dIntS11 in the ICM (gray). The β-CASP domain rotates by 6.3° to create the pocket for binding the C terminus of CG7044. The asterisks indicate the location of the two helices in the CTE.

three independent BRAT1 knockout human HCT116 cell lines and confirmed its complete depletion by western blot analysis and indirect immunofluorescence (Figures 5A and 5B). Interestingly, all three clonal lines displayed no overt growth defects but demonstrated reduced levels of both INTS9 and INTS11 proteins with only a marginal impact on other Integrator subunits (Fig-

ure 5A). This indicates that the INTS9-INTS11-BRAT1 complex is important for the specific stabilization of INTS9/INTS11. Fractionation of the nuclear and cytoplasmic components of the BRAT1 knockout HCT116 cells revealed that the residual levels of INTS11 still localize to the nucleus, indicating that BRAT1 is not required for transport of INTS11 (Figure 5C). RNAi of

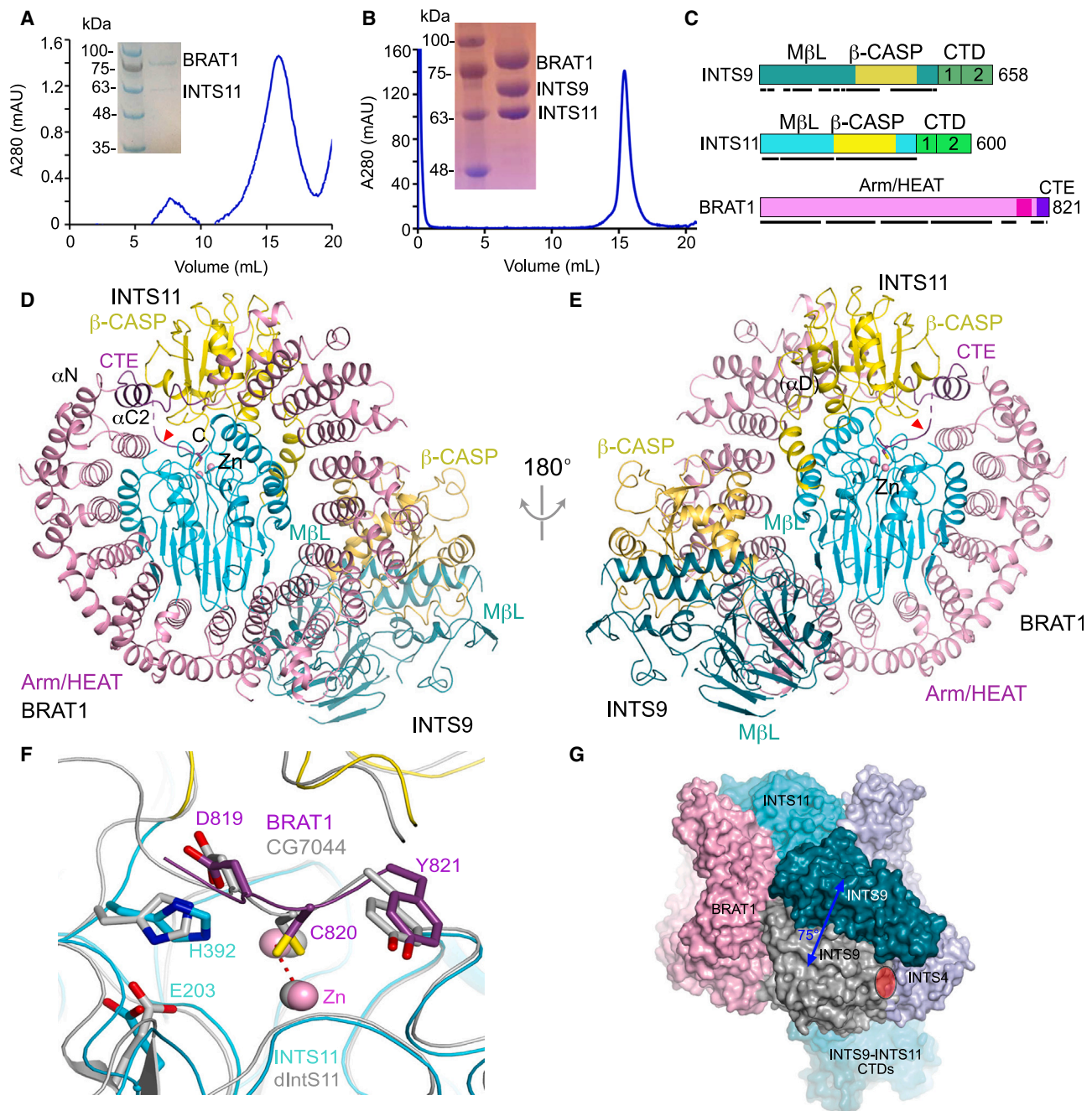


Figure 4. The overall structure of the human INTS9-INTS11-BRAT1 complex

(A) Gel filtration profile of the human INTS11-BRAT1 complex. The expression levels were very low.
(B) Gel filtration profile of human INTS9-INTS11-BRAT1 complex.
(C) Domain organizations of human INTS9, INTS11, and BRAT1. The homologous domains in INTS9 are given slightly darker colors compared with INTS11. The segment in BRAT1 equivalent to the lasso in CG7044 is shown in magenta, but it is disordered.
(D) The overall structure of the human INTS9-INTS11-BRAT1 complex.
(E) The INTS9-INTS11-BRAT1 complex viewed after 180° rotation around the vertical axis. The expected position of helix α D of the β -CASP domain is indicated, but it is disordered in this structure.
(F) Overlay of the binding mode of the C terminus of BRAT1 (violet) in human INTS11 (cyan and yellow) with that of CG7044 in *Drosophila* dlntS11 (gray).
(G) The overall structure of the human INTS9-INTS11-BRAT1 complex.

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CG7044 in *Drosophila* S2 cells also caused co-depletion of dIntS11 but did not overtly affect dIntS9 expression (Figure 5D). These results are consistent with human INTS9 and INTS11 associating with BRAT1, but only dIntS11 forming a stable complex with CG7044.⁵⁵

Next, we examined the functional consequences of BRAT1 knockout and found a striking elevation of misprocessed UsnRNAs (Figure 5E), a hallmark of INTS11-depletion phenotype and Integrator misfunction, and consistent with the overall reduction in INTS9/INTS11 accumulation and previous reports⁴¹ (Figure 5A). The BRAT1 knockout lines created an optimal system to re-express BRAT1 or BRAT1 mutant proteins to test for functional rescue. Specifically, we generated BRAT1 constructs that are either wild-type (WT), have the DCY motif changed to alanine (AAA), have the DCY motif removed (Δ DCY), or lack the CTE, including the α C1 and α C2 helices (Δ CTE). To assess the ability of BRAT1 constructs to rescue the Integrator-depletion phenotype, we utilized a previously described fluorescence-based reporter system where the U7snRNA promoter and coding region are placed upstream of a GFP open reading frame (Figure 5F).^{52,56,57} As expected, transfection of this U7-GFP reporter into control HCT116 cells yielded little GFP expression versus a high amount of GFP produced in BRAT1 null cells (Figure 5F, lanes 1 versus 2). When BRAT1-WT is re-expressed in BRAT1-null cells, we observed low levels of GFP expression, comparable to control HCT116 cells, indicating a complete rescue of the INTS11-depletion phenotype (Figure 5F, lanes 2 versus 3). By contrast, re-expression of the AAA, Δ DCY, or Δ CTE mutant failed to fully restore U7snRNA processing, resulting in GFP production. This indicates that the DCY motif is required for BRAT1 function with INTS11 (Figure 5F). Finally, we also overexpressed INTS11 in BRAT1 knockout cells and found it can almost fully rescue Integrator dysfunction, indicating that destabilization of INTS11 is the major driving factor for the observed reduced Integrator nuclear function (Figure 5G). These data indicate that BRAT1/CG7044 stabilizes INTS11 in the cytoplasm, and the INTS11-BRAT1 complex is required for nuclear Integrator function.

Loss of BRAT1 leads to precocious expression of neurogenesis-driving factors

BRAT1 mutations have been attributed to a spectrum of neurological development disorders connected to epilepsy and seizures.^{41,58–60} To further study the role of BRAT1 in neural differentiation, we generated syngeneic BRAT1 knockout (Δ BRAT1) and control cell lines from human embryonic H7 (WA07) stem cells. Two independent cell lines were generated for both control and Δ BRAT1 human embryonic stem cell (hESC) lines. Δ BRAT1 embryonic stem cells (ESCs) do not exhibit growth defects and exhibit normal ESC morphology and pluripotency marker expression (Figure S6A). To examine the potential effect of BRAT1 loss on the generation of interneuron precursors, we

further differentiated the ESCs into ventral NOs (Figures 6A and S6A–S6C).⁶¹ Similar to the HCT116 cells that lacked BRAT1, the Δ BRAT1 ESCs and NOs both display significant UsnRNA misprocessing, demonstrating that at least at the level of UsnRNA, there is an apparent Integrator deficiency phenotype (Figure 6B).

To determine the impact of BRAT1 depletion on gene expression in hESCs and NOs, we performed RNA sequencing (RNA-seq). We found that Δ BRAT1 hESCs did not distinctly cluster from control hESCs using principal-component analysis (PCA) of the RNA-seq, and consistently, only a limited number of genes exhibited differential expression (Figures S6C and S6D). By contrast, the PCA analysis of the NO RNA-seq data revealed tight clustering of the two Δ BRAT1 clonal lines and separate from that of control lines (Figure 6C). Further, unlike hESCs, the Δ BRAT1 neural organoids displayed a much more disrupted transcriptome (Figures 6D and 6E). We validated several of the observed changes using RT-qPCR (Figure S6E). Gene Ontology analysis of the downregulated genes did not identify any processes above the statistical cutoff for significance. By contrast, the upregulated genes were enriched in many pathways involved in neurogenesis, axon formation, and neurodifferentiation (Figure 6F). Further, many top upregulated genes are considered master-regulator transcription factors (DLX1, DLX2, DLX5, and NKX2-1) and are implicated in brain patterning and development.^{62–68} Altogether, these data reveal that loss of BRAT1 disrupts gene expression patterns during neural differentiation.

UBE3D may be a “BRAT1-like” binding partner for CPSF73

Inspired by our observations on the INTS11-BRAT1 complex, we wondered whether the INTS11 paralog, CPSF73, could have a binding partner with a similar interaction mechanism. To explore this possibility, we first used the BRAT1 structure to search against all the human protein structures predicted by AlphaFold2,⁶⁹ but failed to identify a structurally similar protein using Dali.⁷⁰ We then screened through the list of binding partners of human CPSF73 in the BioGRID database,⁷¹ using AlphaFold-Multimer⁷² to predict the structure of its complex with the CPSF73-CPSF100 heterodimer. This led to the identification of a complex between CPSF73 and UBE3D (ubiquitin protein ligase 3D). UBE3D primarily interacts with the metallo- β -lactamase and β -CASP domains of CPSF73 and has essentially no contacts with CPSF100. A prediction for the CPSF73-UBE3D binary complex produced a similar model (Figures 7A and S7A), as well as another model where there are also some contacts between UBE3D and the CTDs of CPSF73 (Figure S7B). Most interestingly, both models suggest that Cys144 at the tip of a long segment (residues 129–159) that extends far from the body of UBE3D is positioned in the active site of CPSF73 and can directly coordinate one or both metal ions (Figure 7B). CPSF73 is in a fully open conformation, with a canyon between the

(G) Overlay of the structures of the ICM and the INTS9-INTS11-BRAT1 complex, based on the metallo- β -lactamase domain of INTS11. INTS11: cyan; INTS9 in the ICM: dark cyan; INTS9 in the INTS9-INTS11-BRAT1 complex: gray; INTS4: light blue; BRAT1: pink. The red oval indicates clashes between INTS9 in the INTS9-INTS11-BRAT1 complex and INTS4 in the ICM, while the blue arrow indicates large differences in the position and orientation of INTS9 between the two complexes.

See also Video S1.

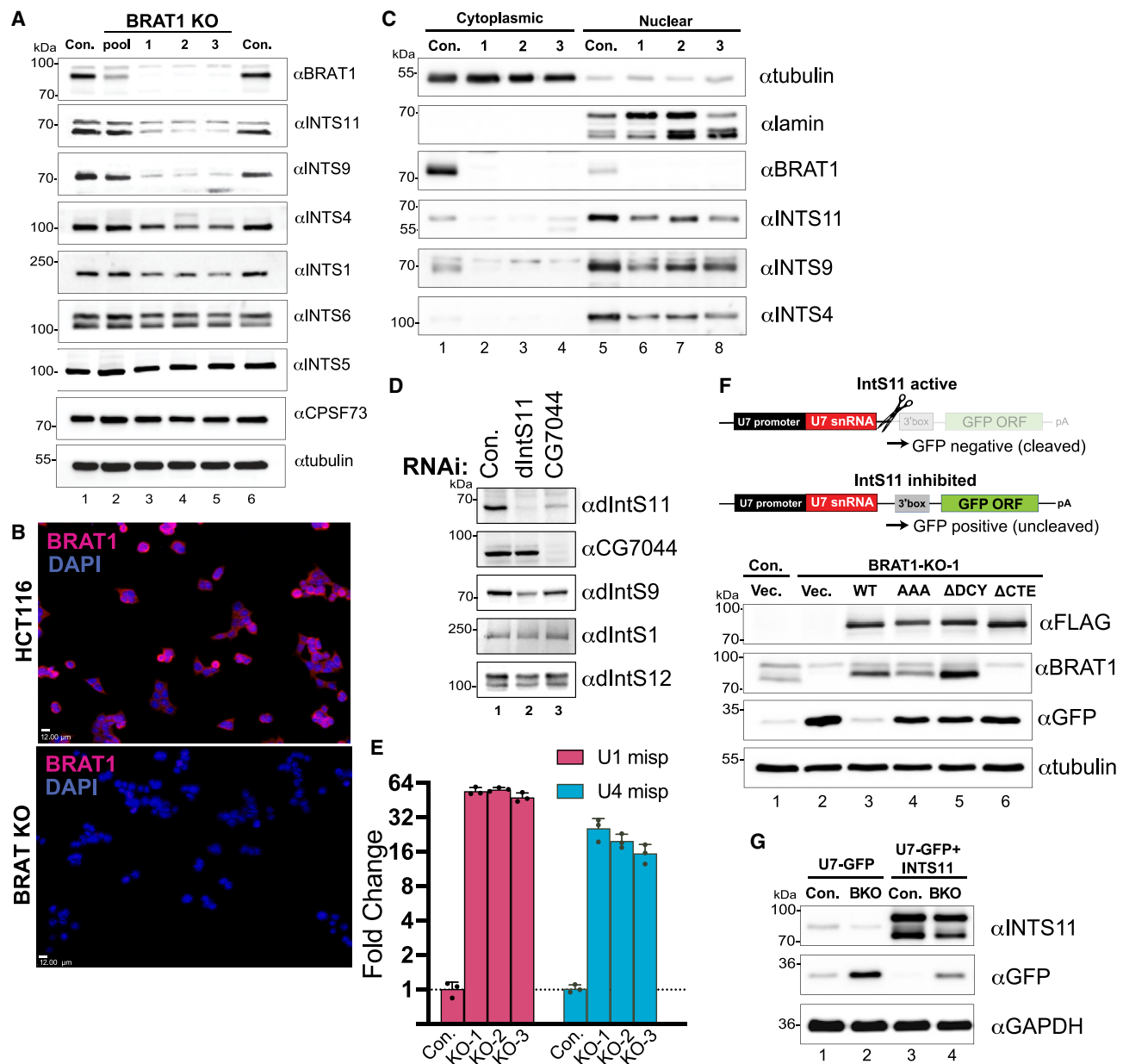


Figure 5. BRAT1 stabilizes INTS11 in the cytoplasm and is required for Integrator function

(A) Western blot probing for HCT116 control line (Con.), a pool cell line derived from transfection of HCT116 cells with sgRNA targeting BRAT1 (pool), and 3 independent HCT116 BRAT1 knockout clonal lines (BRAT1 KO-1, -2, and -3).

(B) Immunofluorescence of HCT116 cells (WT) and HCT116 BRAT1 knockout line with nuclei labeled with DAPI (blue) and BRAT1 labeled with TX-Red (red). Scale bar indicates 12 μ m.

(C) Western blot analysis of cytoplasmic and nuclear fractions of WT HCT116 cells (Con.) and the 3 BRAT1 knockout lines (BRAT1 KO-1, -2, and -3).

(D) Western blot of lysates from *Drosophila* S2 cells treated with dsRNA targeting either dlIntS11, CG7044, or a non-specific target (Con.).

(E) RT-qPCR specific for misprocessed U1snRNA (U1 misp) or U4snRNA (U4 misp). Data are quantified from triplicate RNA isolations, normalized to 7SK RNA, and plotted as fold change relative to control.

(F) Top, schematic of the U7-GFP reporter that produces low levels of GFP protein when Integrator is functional but produces significant GFP expression due to transcriptional readthrough when Integrator function is diminished. Bottom, western blot analysis of cell lysates derived from HCT116 cells (Con.) transfected with the U7-GFP reporter or from BRAT1 KO-1 cells also transfected with the reporter and with BRAT1 cDNA. Note that the BRAT1 cDNA was either WT, had the DCY changed to AAA (AAA), lacked the DCY (Δ DCY), or had the CTE deleted (Δ CTE).

(G) Western blot analysis of HCT116 cells (Con.) or BRAT1 KO-1 transfected with the U7-GFP reporter or the U7-GFP reporter and INTS11.

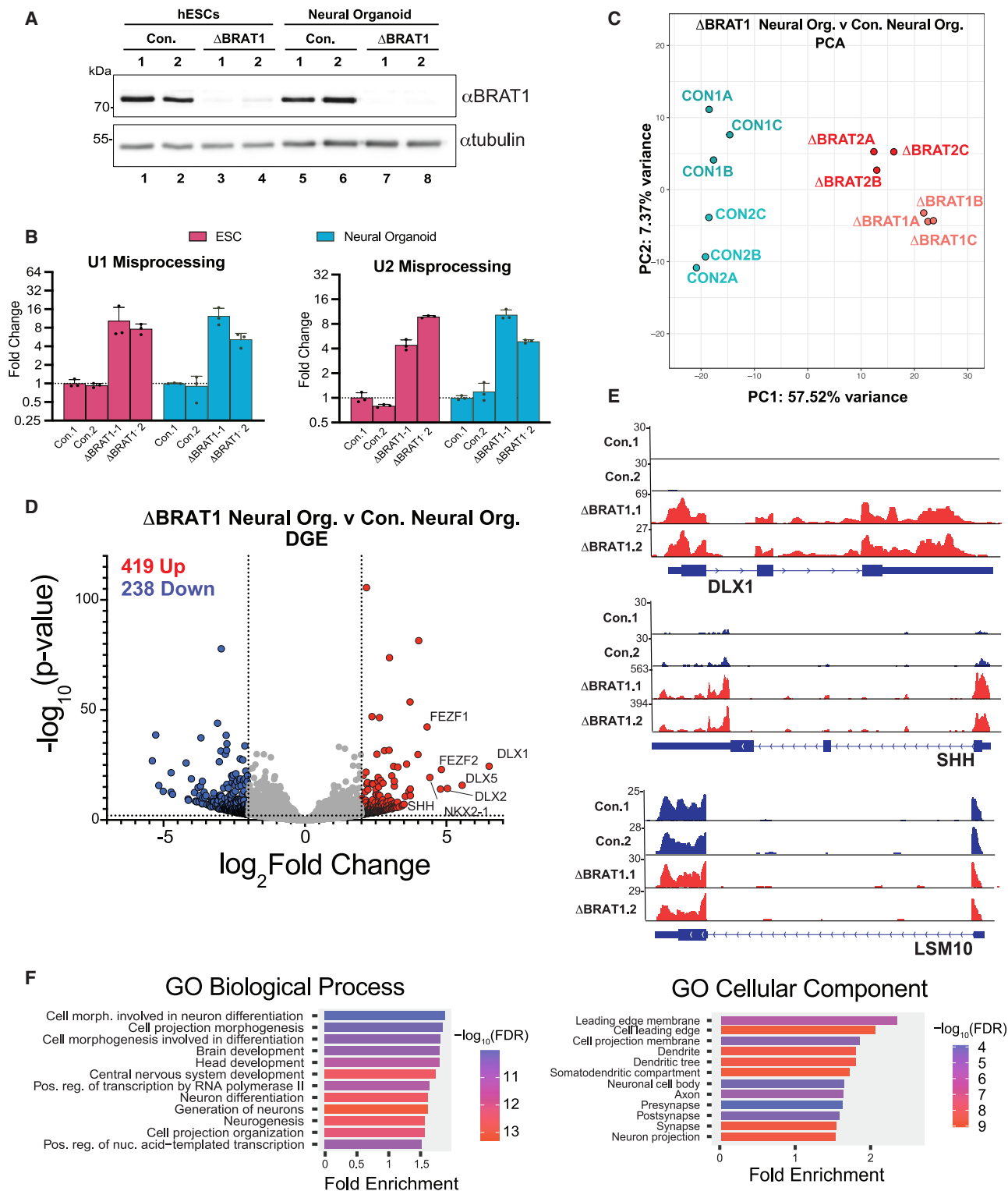


Figure 6. BRAT1 is required for transcriptomic integrity during neural differentiation

(A) Western blot of lysates derived from human embryonic stem cells (hESCs) or neural organoids derived from hESCs. Expression of BRAT1 was analyzed in two independent clones each of syngeneic control (Con.) or BRAT1 knockout (Δ BRAT1) cell lines. Lanes 1–4: undifferentiated embryonic stem cells. Lanes 5–8: neural organoids derived from control and Δ BRAT1 hESCs.

(legend continued on next page)

metallo- β -lactamase and β -CASP domains to accommodate this segment from UBE3D (Figure S7C). Cys144 and the following His145 are conserved among UBE3D homologs, including the yeast homolog Ipa1 (Figure 7C), suggesting their functional importance.

To gain cellular evidence that UBE3D binds CPSF73, we first probed the localization of several CPA factors and UBE3D. As was the case for BRAT1, UBE3D is primarily cytoplasmic, whereas CPSF73 and CPSF100 were more enriched in the nucleus but still demonstrate cytoplasmic localization (Figure 7D). In addition, we verified the importance of that long segment of UBE3D for CPSF73 binding. We generated three mutants, deleting residues 131–156 or 135–154 of the segment or replacing the WCCH sequence (Figure 7C) at the tip of the segment with alanines (142–145A). We observed that all UBE3D FLAG-tagged proteins expressed to comparable levels, but all three mutants were found to pull down significantly less HA-CPSF73 (Figure 7E). We co-expressed the yeast *Kluyveromyces lactis* Ysh1 (CPSF73 homolog), Ydh1 (CPSF100 homolog), Pta1 (symplekin homolog), and Ipa1 in insect cells and successfully purified a complex containing Ysh1 and Ipa1 (Figure S7D). AlphaFold2 prediction of the Ysh1-Ipa1 complex showed similar overall structures and conserved modes of interaction in the active site region (Figures S7E and S7F). We generated equivalent mutations in *K. lactis* Ipa1 and showed that the mutants could no longer co-purify with Ysh1 after co-expression in insect cells (Figure 7F). Altogether, these results indicate that UBE3D associates with only CPSF73 in the CPA machinery and that the long segment is important for the binding.

DISCUSSION

Here, we have revealed binding partners that stabilize INTS11 and CPSF73 in the cytoplasm, and these interactions are important for their functions in the nucleus. Therefore, the partners behave like chaperones of these endonucleases in the cytoplasm. Our studies also suggest striking similarities in the mechanisms of these chaperone partners. They both occupy the active site of their target endonuclease and present a conserved cysteine residue to directly coordinate the metal ions. The levels of the endonucleases are greatly reduced in the absence of the partners, leading to RNA misprocessing and other functional defects. This suggests that a chaperone-mediated regulation of endonuclease activity and subcellular localization is a conserved mechanism by which the function of critical RNA processing machinery is controlled.

The location of the BRAT1 C terminus in the active site of INTS11 demonstrates that BRAT1 should inhibit the nuclease

activity of INTS11 (Figure 4G). Further, the organization of INTS11 and INTS9, while associated with BRAT1, would prevent their association with INTS4 and the formation of the ICM (Figures 3 and 4). These structural insights appear at odds with the INTS11 loss-of-function phenotype observed in BRAT1 knockout cells (Figures 5 and 6). A parsimonious model (Figure 7G) consistent with both data would be that BRAT1 is a chaperone partner of newly synthesized INTS11, stabilizing this endonuclease in the cytoplasm before its association with the nuclear INT. While INTS11 alone is expected to exist in a closed, inactive state, its conformations may be more dynamic, and BRAT1 could ensure no spurious endonucleolytic events. These data also suggest that INTS11 exists in two separate cellular complexes: those bound to the full nuclear INT, ready to function in transcriptional regulation, and those in complexes with BRAT1 (and INTS9). This cytoplasmic pool of INTS11 likely represents the early stages of an ICM assembly pathway and possibly also forms a reservoir of this enzyme in the cytoplasm that becomes necessary under specific cell states (e.g., neurogenesis).

Both homozygous and compound heterozygous recessive mutations in the *BRAT1* gene have been found in patients.^{47,48} Our structure of BRAT1 provides a foundation for understanding the molecular basis of how these mutations could perturb its function (Figure S5I; Table S2). Many of these mutations are expected to disrupt the structure of BRAT1, which will likely lead to its destabilization. This is supported by the observation that the V62E mutation greatly reduced the levels of BRAT1 protein in cells.⁴¹ Only two of these BRAT1 missense mutations are located in the interface with INTS11, showing ion-pair or hydrogen-bonding interactions, while none are in the interface with INTS9.

Our data further reveal that BRAT1 is surprisingly not essential in HCT116 or hESCs and that persistent reduction of INT function can be tolerated (Figure 5). However, upon differentiation of ESCs to neural organoids, a requirement for BRAT1 is uncovered as these cells demonstrate precocious expression of several key transcription factors and signaling molecules that drive neurogenesis (Figure 6). These observations are consistent with INT functioning as a transcriptional attenuator and suggest that patients with BRAT1 mutations may have disrupted transcriptome remodeling critical for neuronal development. In particular, using a ventrally fated neural organoid induction model, we found upregulation of SHH, NKX2-1, DLX1, DLX5, and DLX2, which points toward dysregulation of interneuron specification.^{62–68,73} This imbalance may contribute to the pathology observed in patients with BRAT1 mutations, such as those with RMFSL.

UBE3D and Ipa1 are poorly understood, but it was recently reported that they may interact with CPSF73/Ysh1 and stabilize

(B) Results from RT-qPCR from cells shown in (A) for U1 snRNA or U2 snRNA misprocessing. Measurements were conducted in biological replicates ($n = 3$) with error bars representing standard deviation.

(C) Principal-component analysis plot generated from DESeq2 analysis of RNA-seq on Δ BRAT1 versus control neural organoids, corresponding to samples from lanes 5–8 of (A). Each cell line was tested as three independent biological replicates consisting of 20 pooled organoids each.

(D) Volcano plot derived from RNA-seq data collected from the neural organoids shown in (A). Each cell line was tested as three independent biological replicates consisting of 20 pooled organoids each (see also Figure 6C). The significance of differential gene expression (DGE) was set at >4 -fold and adjusted $p < 0.01$.

(E) Gene tracks of selected genes from the RNA-seq data in (D). DLX1 (distal-less homeobox 1) and SHH (sonic hedgehog signaling molecule) were upregulated, and Lsm10 (U7 small nuclear RNA-associated Sm-like protein) was unchanged upon BRAT1 knockout.

(F) Gene Ontology analysis of the differential gene expression in (D) for both biological processes and cellular components.

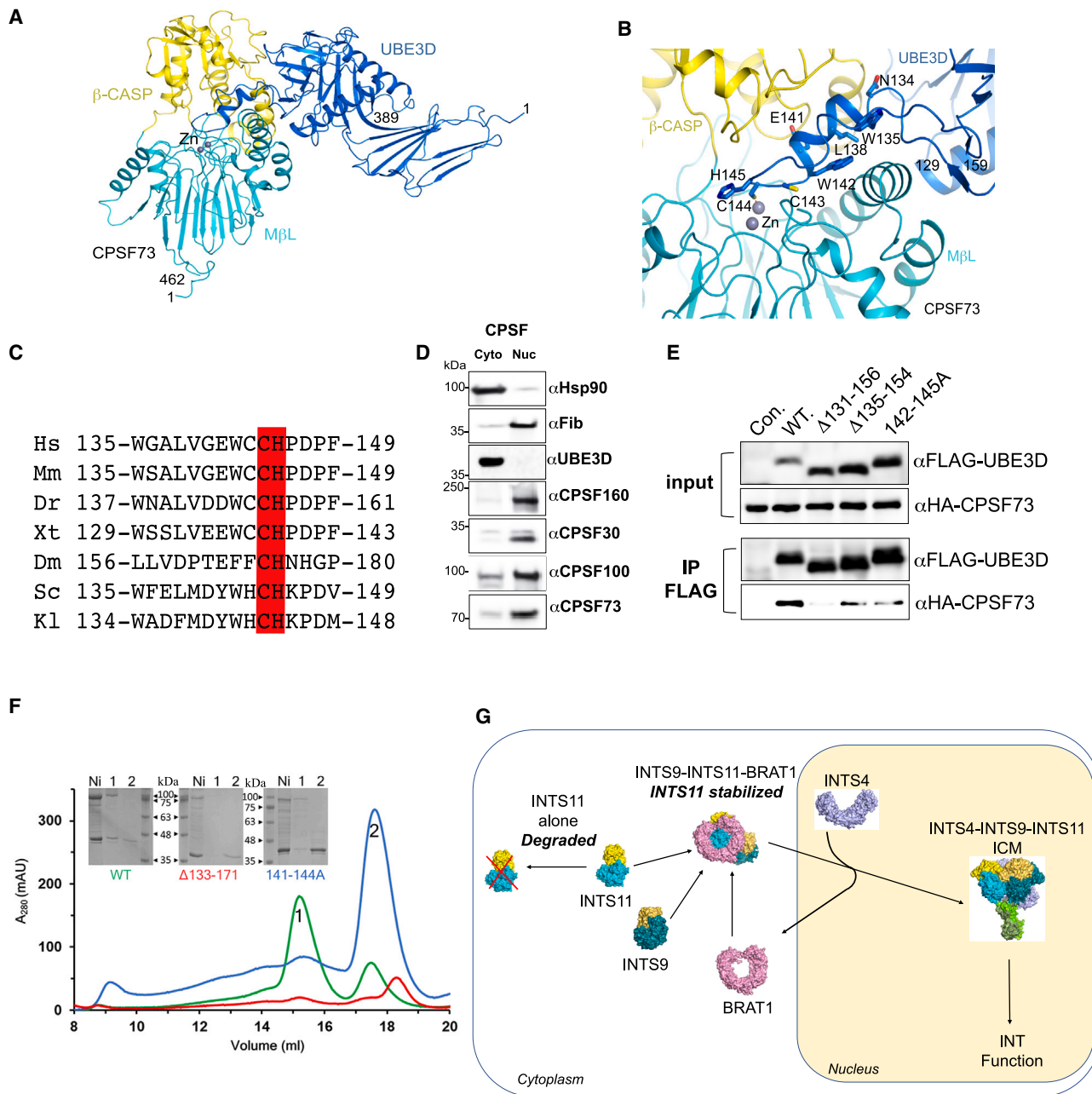


Figure 7. UBE3D is a binding partner of CPSF73 in the cytoplasm

(A) Structure of the human CPSF73-UBE3D complex predicted by AlphaFold-Multimer. UBE3D is shown in marine color.

(B) Detailed interactions between UBE3D and the active site region of CPSF73.

(C) Sequence alignment of selected UBE3D/Ipa1 homologs for the segment in contact with CPSF73 active site. Conserved residues are highlighted in red. Sc, *S. cerevisiae*; Kl, *K. lactis*.

(D) Fractionation of HEK293T cells and western blot analysis indicating primarily cytoplasmic localization of endogenous UBE3D.

(E) Western blot analysis of co-immunoprecipitation with anti-FLAG affinity resin from cell lysates derived from HEK293T cells co-transfected with empty vector or FLAG-tagged UBE3D WT and mutant constructs as well as HA-CPSF73.

(F) Gel filtration profiles of co-expressed WT *K. lactis* Ysh1-Ipa1 complex (green), Ysh1-Ipa1 mutant replacing residues 141–144 with Ala (141–144A, blue), and Ysh1-Ipa1 mutant missing the segment in the CPSF73 active site (red). Inset: SDS gels of the samples.

(G) A model for the role of BRAT1 in Integrator function.

this endonuclease, and reduction in their expression leads to a CPSF73-depletion phenotype.^{74–77} These characteristics are remarkably similar to those described here for BRAT1. Moreover, it is interesting that the binding of UBE3D, annotated as a ubiquitin ligase, to CPSF73 prevents ubiquitination and degradation by the proteasome.^{75,77}

Our work reveals a previously unknown aspect of INTS11 and CPSF73 biogenesis in that chaperones stabilize and engage the active site of these crucial RNA endonucleases. We note that within this same issue, the Jonas laboratory reports similar observations on the role of BRAT1 in Integrator function.⁷⁸ While the two studies are remarkably harmonious, the other report also reveals that IP₆ is important for the conversion from the INTS9-INTS11-BRAT1 complex to the ICM. Overall, these structures further show a compelling form of molecular mimicry where amino acids function as uncleavable substrates reminiscent of RNA targets to ultimately deliver these enzymes to their respective complexes.

Limitations of the study

One potential limitation of our work pertains to insight as to how the transport of ICM subunits into the nucleus is accomplished. Our biochemical approaches, while robust, did not detect any of the well-known import factors. This may indicate that their association may be too transient for our relatively rigorous purification schemes. Another limitation of our work relates to how directly applicable our findings are to human diseases associated with BRAT1 mutations since our neural organoids were derived from hESCs that were null for BRAT1. It is likely that BRAT1 mutations lead to more subtle defects in neuronal development versus a complete knockout, and the effects of BRAT1 deficiency on specific neuronal lineages at later stages of development remain to be evaluated. Finally, our molecular insights on the CPSF73-UBE3D complex are based on predictions by AlphaFold2, and experimentally determining the structure of this complex will be important for the verification of the model, which may reveal additional insights on this interaction.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **RESOURCE AVAILABILITY**
 - Lead contact
 - Materials availability
 - Data and code availability
- **EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**
 - Cell lines
- **METHOD DETAILS**
 - Plasmid construction and stable cell lines generation
 - Nuclear extract preparation
 - Western blotting and anti-FLAG affinity purification
 - Antibody generation and purification
 - Mass spectrometry sample digestion
 - Mass spectrometry analysis
 - Database searching for mass spectrometry
 - Immunofluorescence staining
 - Image acquisition for BRAT1 expression and localization immunofluorescence

- Image analysis for BRAT1 expression and localization immunofluorescence
- Protein expression and purification
- EM specimen preparation and data collection
- Image processing
- Model building
- AlphaFold prediction
- CRISPR genomic editing
- RNA interference (*Drosophila* cells)
- RT-qPCR
- Reporter cell line establishment
- Embryonic stem cell line establishment and organoid differentiation
- Stem cell staining
- Small-scale extract and co-immunoprecipitation
- **QUANTIFICATION AND STATISTICAL ANALYSIS**
 - RT-qPCR quantification and analysis
 - RNA-seq library generation and mapping
 - Genomic statistical tests

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.molcel.2024.06.017>.

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AUTHOR CONTRIBUTIONS

M.K.J., N.D.E., M.H., A.C.B., K.-L.H., and E.J.W. designed and performed biochemical experiments. M.-H.L., H.-F.C., and L.T. conducted all structural analyses. M.K.J. and E.J.W. designed and performed RNA-seq experiments. W.K.R. conducted mass spectrometry. K.W. and C.P. generated ESC lines and organoids. W.C. imaged and processed immunofluorescence. M.K.J., E.J.W., L.T., and C.P. wrote and edited the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
FLAG-HRP	Sigma #A8592	AB_439702
Drosophila IntS1	Ezzeddine et al. ⁵⁷	N/A
Drosophila IntS4	Huang et al. ⁶	N/A
Drosophila IntS6	This paper	N/A
Drosophila IntS7	Huang et al. ⁶	N/A
Drosophila IntS8	Huang et al. ⁶	N/A
Drosophila IntS9	Ezzeddine et al. ⁵⁷	N/A
Drosophila IntS11	Ezzeddine et al. ⁵⁷	N/A
Drosophila IntS12	Chen et al. ⁵⁶	N/A
Hsp90	Abclonal #A0365	AB_2757150
Fibrillarin	Abclonal #A1136	AB_2758523
BRAT1	AbClonal #A12801	AB_2759640
hINTS1	Bethyl #A300-361A	AB_2127258
hINTS4	Bethyl #A301-296A	AB_937909
hINTS5	Proteintech #14069-1-AP	AB_2296187
hINTS9	Proteintech #11657-1-AP	AB_2127514
hINTS11	Abcam #ab75276	AB_937779
Gapdh	ThermoFisher #MA5-15738	AB_10977387
Lamin A/C	Santa Cruz Biotechnology #sc-7292	AB_627875
CG7044	This paper	N/A
hINTS6	Bethyl #A301-658A	AB_1210927
CPSF73	Abclonal #A2222	Pending
tubulin	Abcam #ab15246	AB_301787
GFP	Abclonal #AE100	Pending
UBE3D	Abnova #PAB21883	AB_10966255
CPSF1/CPSF160	Abclonal #A17144	AB_2769032
CPSF4/CPSF30	Abclonal #A10284	AB_2757826
CPSF100	Abclonal #A9297	AB_27692033
HA-HRP	Abclonal #AEO25	AB_2769866
OCT-4	Cell Signaling Technologies #2750	AB_823583
SOX2	Cell Signaling Technologies #3579	AB_2195767
NANOG	Cell Signaling Technologies #4903	AB_10559205
SSEA4	Abcam #ab16287	AB_778073
Tra-1-81	EMD Millipore #MAB4381	AB_177638
Tra-1-60	Invitrogen #MA1-0231000	AB_2536699
Goat anti-rabbit IgG Alexa Fluor 488	Invitrogen #A-11034	AB_2576217
Goat anti-rabbit IgG Alexa Fluor 568	Invitrogen #A-11036	AB_10563566
Goat anti-mouse IgM Alexa Fluor 488	Invitrogen #A-21042	AB_2535711
Goat anti-mouse IgM Alexa Fluor 568	Invitrogen #A-21043	AB_2535712
Goat anti-mouse IgG(H+L) Alexa Fluor 568	Invitrogen #A-11031	AB_144696
Goat anti-rabbit IgG (H+L) Texas Red-X	Invitrogen, #T-6391	AB_2556779

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
AzNTPs	TriLink Technologies	Cat# K-1005
Biotin-11-NTPs	Perkin Elmer	Cat# NEL54(2/3/4/5)001
Critical commercial assays		
NEBNext Poly(A) mRNA Magnetic Isolation Module	NEB	Cat# E7490S
Deposited data		
Raw and analyzed data	This paper	GEO: GSE246833
Raw blots and fluorescence images	This paper	DOI: https://doi.org/10.17632/ygprhc3hvt.1
Structure of the human INTS9-INTS11-BRAT1 complex	This paper	PDB: 8UIB
Structure of the <i>Drosophila</i> dIntS11-CG7044 complex	This paper	PDB: 8UIC
Experimental models: Cell lines		
HEK293T	ATCC	N/A
DL1	Dr. Sara Cherry, UPenn	N/A
S2-DGRC clone 6	DGRC	N/A
S2 Flag-dIntS5	Huang et al. ⁶	N/A
S2 Flag-dIntS11	Elrod et al. ⁴	N/A
S2 Flag-dIntS1	Huang et al. ⁶	N/A
S2 Flag-CG7044	This paper	N/A
S2 Flag-dIntS8	This paper	N/A
HEK293T-TRE-FLAG-INTS11	This paper	N/A
HEK293T-TRE-FLAG-BRAT1	This paper	N/A
HCT116	ATCC	N/A
HCT116-BRAT1KO-1	This paper	N/A
HCT116-BRAT1KO-2	This paper	N/A
HCT116-BRAT1KO-3	This paper	N/A
Human H7 Embryonic Stem Cells (H7-hESC)	WiCell Research Institute	NIHhESC-10-0061
H7-hESC-Con-1	This paper	N/A
H7-hESC-Con-2	This paper	N/A
H7-hESC-DBRAT1-1	This paper	N/A
H7-hESC-DBRAT1-2	This paper	N/A
HEK293T-TRE-FLAG-UBE3D	This paper	N/A
Oligonucleotides		
Table S3	This paper	N/A
Recombinant DNA		
pMT-3xFLAG-puro vector	Elrod et al. ⁴	N/A
pMT-3xFLAG-dIntS5-puro	Huang et al. ⁶	N/A
pMT-3xFLAG-dIntS11-puro	Huang et al. ⁶	N/A
pMT-3xFLAG-dIntS1-puro	Huang et al. ⁶	N/A
pMT-3xFLAG-dIntS5-puro	Huang et al. ⁶	N/A
pMT-3xFLAG-CG7044-puro	This paper	N/A
pMT-3xFLAG-dIntS8-puro	Huang et al. ⁶	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
FastP 0.23.1	Chen et al. ⁷⁹	N/A
STAR_2.7.9a	Dobin et al. ⁸⁰	N/A
featureCounts	Liao et al. ⁸¹	N/A
DESeq2 1.34.0	Love et al. ⁸²	N/A
R v4.0.2	www.R-project.org	N/A
pcaExplorer	Marini and Binder ⁸³	N/A
EnrichR	Chen et al. ⁸⁴	N/A
ggplot2	Wickham ⁸⁵	N/A
IGV	Robinson et al. ⁸⁶	N/A
ShinyGO 0.77	Ge et al. ⁸⁷	N/A
Prism v10.1.0	GraphPad	N/A
Morpheus	Morpheus, https://software.broadinstitute.org/morpheus	N/A

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Liang Tong (ltong@columbia.edu).

Materials availability

Cell lines generated and described in this study are available upon request.

Data and code availability

- The atomic coordinates have been deposited at the PDB (entry codes 8UIB and 8UIC). GEO Accession numbers: all datasets generated in this study are available for download from GEO: GSE246833. All source data is available at Mendeley Data, DOI: <https://doi.org/10.17632/ygprhc3hvt.1>. The deposited data will be publicly available upon publication of the paper.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**Cell lines**

HEK 293T and HCT116 cells were grown with 5% CO₂ in DMEM (Gibco, #11965-092), supplemented with 10% (v/v) FBS and 1% (v/v) penicillin-streptomycin (Gibco, #15070-063) at 37°C. *Drosophila* S2 cells were grown in Sf-900 II SFM (Gibco, #10902-088), supplemented with 1% (v/v) Antibiotic-Antimycotic at 27°C. Human embryonic stem cells (H7, WA07, WiCell) and derivatives (Control-1, Control-2, ΔBRAT1-1, ΔBRAT1-2) were maintained in Stemflex medium (Gibco, #A33494-01) on LDEV-free, growth factor reduced Matrigel basement membrane (Corning #356254) at 37°C in humidified 5%CO₂ and 5%O₂ atmosphere.

METHOD DETAILS**Plasmid construction and stable cell lines generation**

Flag-tagged *Drosophila* Integrator subunits and CG7044 were cloned into the pMT-3xFLAG-puro vector⁴ to inducibly express in S2 cells. Plasmids were all sequenced to confirm identity. *Drosophila* S2 cells stably expressing the indicated Flag-tagged Integrator subunits or CG7044 were generated by plating 2x10⁶ cells in a 6-well dish in complete media. After 1 hour, 2 μg of plasmids were transfected using Fugene HD (Promega, #E2311). The next day, 2.5 μg/mL puromycin was added to select and maintain the cell population.

Human Flag-tagged Integrator subunits or BRAT1 were generated by cloning cDNAs into the doxycycline inducible pTRE-3xFLAG-Blast.⁴ HEK 293T cell lines expressing the plasmids were generated by plating 4x10⁵ cells in a 24-well dish in complete media. Cells were transfected with 800 μg plasmid using Lipofectamine 2000 (Invitrogen, #11668027). After 24 hours, 10 μg/mL blasticidin was added to the media to select the cell population, and for maintenance.

Nuclear extract preparation

Nuclear and cytoplasmic extracts were performed using a variation of the Dignam method.⁸⁸ Briefly, HEK 293T or *Drosophila* S2 cells were grown into nearly confluent 150 mm dishes, collected, and rinsed with cold PBS. Prior to collection, HEK 293T cells were pretreated with doxycycline for 24 hrs and *Drosophila* cells were pretreated with 500 μ M CuSO₄ for 24 hours to induce protein expression. The cells were resuspended in five times the pellet volume of Buffer A (10mM Tris-HCl pH7.9, 1.5mM MgCl₂, 10mM KCl, 0.5mM DTT, and 0.2mM PMSF) and incubated at 4°C for a 15-minute rotation to allow cells to swell. Cells were pelleted down at 800g for 10 minutes at 4°C, then resuspended in two times the pellet volume of Buffer A and homogenized on ice with dounce pestle B for 30 gentle strokes before centrifugation at 2,000g for 10 minutes at 4°C. From here, the supernatant containing the cytoplasmic fraction is collected. The nuclear fraction in the pellet was resuspended and washed gently with a 5-minute rotation two times in Buffer D (20mM Na HEPES pH7.4, 100mM KCl, 0.2mM EDTA, and 0.5mM DTT) and one time in Buffer A. The sample was again homogenized with dounce pestle B for 15 strokes before rotating for 60 minutes at 4°C. Sample was centrifuged at 15,000g for 30 minutes at 4°C. Supernatant was collected and both the nuclear and cytoplasmic fractions were subjected to overnight dialysis at 4°C in 2 liters Buffer D (20mM Na HEPES, 100mM KCl, 0.2mM EDTA, 0.5mM DTT, and 20% glycerol). To remove any precipitate before further analysis, extracts were centrifuged again at 15,000g for 3 minutes at 4°C.

Western blotting and anti-FLAG affinity purification

Cells were lysed to collect protein by directly adding 2X SDS sample buffer (120mM Tris-HCl pH6.8, 4% SDS, 200mM DTT, 20% Glycerol, and 0.02% Bromophenol blue) to cells in wells. Samples were then incubated at room temperature with periodic swirling before a 10-minute boiling at 95°C and a short sonication. Denatured protein samples were then resolved on a 10% SDS-PAGE gel and transferred to a PVDF membrane (Bio-Rad, #1620177). For both commercial and custom-designed antibodies, blots were probed as previously described.⁴

Blots were probed by custom-designed *Drosophila* antibodies as previously described⁴ diluted in PBS-0.1% Tween supplemented with 5% nonfat milk. To detect proteins from 293T lysate, anti-hInts11 (Bethyl, #A301-274A), anti-hIMP1 (Thermo, #PA5-21629), anti-GFP (Clontech, #632381), anti- α Tubulin (abcam, #ab15246), and anti-GAPDH (Thermo, #MA5-15738) were used at the dilution suggested by the manufacturer.

To purify FLAG-tagged Integrator complexes, 500 μ g of nuclear extract was mixed with 30 μ l anti-Flag M2 affinity agarose slurry (Sigma, #A2220) equilibrated in binding buffer (20mM Na HEPES pH7.4, 150mM KCl for *Drosophila*, 100mM KCl for human, 10% Glycerol, 0.1% NP-40) and rotated for 4 hours at 4°C. Next, four washes were performed out in binding buffer by rotating at 4°C for 5 minutes followed by a 500g centrifugation at 4°C. After the final wash, the supernatant was removed, 50 μ l of 2X sample buffer was added to the anti-FLAG resin to elute the proteins, and the samples boiled at 95°C for ten minutes. To generate western blot input samples, equal volume of 2X loading buffer was added to nuclear or cytoplasmic extracts and 1/10 of the immunoprecipitation was loaded as estimated by protein mass.

Antibody generation and purification

Commercial antibodies used to detect proteins are listed in the [key resources table](#). Others were custom-made and were raised against recombinant proteins expressed in *E. coli* as described previously.^{4,6} Specifically, these proteins were: the first 100 amino acids of the N-termini of either *Drosophila* IntS4, *Drosophila* IntS7, or *Drosophila* IntS8. These recombinant proteins were then shipped to a commercial vendor (Cocalico Biologicals, PA) and used to inoculate guinea pigs. Sera was isolated and tested for specific reactivity to target proteins using western blot analysis initially with nuclear extract to confirm that protein bands matched predicted size. Sera passing this filter was subsequently tested by probing lysates from DL1 cells treated with dsRNA targeting the protein of recognition to confirm loss of specific bands.

Mass spectrometry sample digestion

Drosophila samples were prepared at the University of Texas Medical Branch Mass Spectrometry Facility similar to what was described previously.⁸⁹ Briefly, the agarose bead-bound proteins were washed several times with 50mM Triethylammonium bicarbonate (TEAB) pH 7.1, before being solubilized with 40 μ L of 5% SDS, 50mM TEAB, pH 7.55 followed by a room temperature incubation for 30 minutes. The supernatant containing the proteins of interest was then transferred to a new tube, reduced by making the solution 10mM Tris(2-carboxyethyl)phosphine (TCEP) (Thermo, #77720), and further incubated at 65°C for 10 minutes. The sample was then cooled to room temperature and 3.75 μ L of 1M iodoacetamide acid was added and allowed to react for 20 minutes in the dark after which 0.5 μ L of 2M DTT was added to quench the reaction. Then, 5 μ L of 12% phosphoric acid was then added to the 50 μ L protein solution followed by 350 μ L of binding buffer (90% Methanol, 100mM TEAB final; pH 7.1). The resulting solution was administered to an S-Trap spin column (Protifi, Farmingdale NY) and passed through the column using a bench top centrifuge (30 second spin at 4,000g). The spin column was then washed three times with 400 μ L of binding buffer and centrifuged (1200rpm, 1min). Trypsin (Promega, #V5280, Madison, WI) was then added to the protein mixture in a ratio of 1:25 in 50mM TEAB, pH=8, and incubated at 37°C for 4 hours. Peptides were eluted with 80 μ L of 50mM TEAB, followed by 80 μ L of 0.2% formic acid, and finally 80 μ L of 50% acetonitrile, 0.2% formic acid. The combined peptide solution was then dried in a speed vacuum (room temperature, 1.5 hours) and resuspended in 2% acetonitrile, 0.1% formic acid, 97.9% water and aliquoted into an autosampler vial.

Human samples were prepared at the University of Rochester Mass Spectrometry Resource Lab. To allow total protein to be evaluated in a single digest, FLAG immunoprecipitated samples were run on a 4–12% SDS-Page gel to remove contaminants and create a region about 10mm long. Each region was excised from the gel and cut into smaller cubes, about 1mm in size. Samples were destained, reduced with DTT, alkylated with IAA (Sigma), and dehydrated with acetonitrile. Dehydrated gel pieces were incubated at room temperature for half an hour after the addition of trypsin at 10 ng/ μ L in 50 mM ammonium bicarbonate to just cover the pieces. After the room temperature incubation, the pieces were completely submerged by the addition of more ammonium bicarbonate, and incubated at 37°C overnight. The next day, 50% acetonitrile and 0.1% TFA were used to extract the peptides, which were then dried down in a CentriVap concentrator (Labconco). Finally, the peptides were desalted with homemade C18 spin columns, dried once more, and reconstituted in 0.1% TFA.

Mass spectrometry analysis

Drosophila peptide mixtures were analyzed by nanoflow liquid chromatography-tandem mass spectrometry (nanoLC-MS/MS) using a nano-LC chromatography system (UltiMate 3000 RSLCnano, Dionex), coupled on-line to a Thermo Orbitrap Eclipse mass spectrometer (Thermo Fisher Scientific, San Jose, CA) through a nanospray ion source. A direct injection method is used onto an analytical column; Aurora (75 μ m X 25 cm, 1.6 μ m) from (ionopticks). After equilibrating the column in 98% solvent A (0.1% formic acid in water) and 2% solvent B (0.1% formic acid in acetonitrile (ACN)), the samples (2 μ L in solvent A) were injected (300 nL/min) by gradient elution onto the C18 column as follows: isocratic at 2% B, 0–10 min; 2% to 27% 10–98 min, 27% to 45% B, 98–102 min; 45% to 90% B, 102–103 min; isocratic at 90% B, 103–104 min; 90% to 15%, 104–106 min; 15% to 90% 106–108 min; isocratic for two minutes; 90%–2%, 110–112 min; and isocratic at 2% B, till 120 min.

All *Drosophila* LC-MS/MS data were acquired using an Orbitrap Eclipse in positive ion mode using a top speed data-dependent acquisition (DDA) method with a 3 second cycle time. The survey scans (m/z 375–1500) were acquired in the Orbitrap at 50,000 resolution (at m/z = 400) in profile mode, with a maximum injection time of 100 ms and an AGC target of 400,000 ions. The S-lens RF level was set to 30. Isolation was performed in the quadrupole with a 1.6 Da isolation window, and HCD MS/MS acquisition was performed in profile mode using the orbitrap at a resolution of 7500 using the following settings: parent threshold = 5,000; collision energy = 30%; using the default settings. Monoisotopic precursor selection (MIPS) and charge state filtering were on, with charge states 2–6 included. Dynamic exclusion was used to remove selected precursor ions, with a $\pm$ 10 ppm mass tolerance, for 60 seconds after acquisition of one MS/MS spectrum.

Prepared human peptides were injected onto a homemade 30 cm C18 column with 1.8 μ m beads (Sepax) using an Easy nLC-1200 HPLC (Thermo Fisher) coupled with a Fusion Lumos Tribrid mass spectrometer (Thermo Fisher). Solvent A consisted of 0.1% formic acid in water, and solvent B consisted of 0.1% formic acid in 80% acetonitrile. A Nanospray Flex source operating at 2 kV was used to introduce ions to the mass spectrometer. The gradient eluted as follows: isocratic at 3% B, 0–2 min; 3% to 10% B in 3–9 min; 10% to 38% B in 10–45 min; 38% to 90% B, 46–51 min; isocratic at 90%, 52–55 min; 90% to 3% B, 56–58 min; and isocratic at 3% B, 59–65 min. The Fusion Lumos was in data-independent mode. The MS1 scan was done over a 390–1010 m/z range with a 60,000 resolution at 200 m/z . The AGC target was 4e5 AGC, and a 50 ms maximum injection time was used. A staggered windowing scheme of 20 m/z with 10 m/z overlaps was used to measure precursor ions. The first cycle fragmented ions between 400–420 m/z , 420–440 m/z , and so on, until the final window of 980–1000 m/z . The second cycle fragmented 390–410 m/z , 410–430 m/z , and so on, until the final window of 990–1010 m/z . There were 30 MS2 scans per cycle, and fragment ions were collected between 200–2000 m/z . Higher energy C-trap dissociation (HCD) was used to fragment precursor ions with a 33% collision energy. A resolution of 15,000, AGC target of 4e5, and a maximum injection time of 23 ms were used for MS2 scans collected in the orbitrap.

Database searching for mass spectrometry

For *Drosophila* samples, tandem mass spectra were extracted and charge state deconvoluted using Proteome Discoverer (Thermo Fisher, version 2.2.0388). Deisotoping was not performed. All MS/MS spectra were searched against a Uniprot *Drosophila* database (version 04-04-2018) using Sequest. Searches were performed with a parent ion tolerance of 5 ppm and a fragment ion tolerance of 0.60 Da. Trypsin was specified as the enzyme, allowing for two missed cleavages. Fixed modification of carbamidomethyl (C) and variable modifications of oxidation (M) and deamidation were specified in Sequest. Representative heatmaps of normalized total spectral counts were generated using Morpheus (<https://software.broadinstitute.org/morpheus>).

For human samples, raw data was processed with DIA-NN version 1.8.1 (<https://github.com/vdemichev/DIA-NN>) in library-free analysis mode.⁴ The human UniProt ‘one protein sequence per gene’ database with ‘deep learning-based spectra and RT prediction’ was used to annotate the library (UP000005640_9606, downloaded 4/7/2021). For precursor ion generation, the maximum number of missed cleavages was set at 1, maximum number of variable modifications at 1 for Ox(M), peptide length range at 7–30, precursor charge range at 2–3, precursor m/z range at 400–1000, and fragment m/z range at 200–2000. To quantify, the ‘Robust LC (high precision)’ mode was used with RT-dependent median-based cross-run normalization and MBR enabled. Protein inferences was set to ‘Genes’ and ‘Heuristic protein inference’ was off. The software automatically set MS1 and MS2 mass tolerances and window size. Library precursor q-value (1%), library protein group q-value (1%), and posterior error probability (50%) were used to filter precursors. The MaxLQ algorithm was used to carry out protein quantification as a part of the DIA-NN R package (<https://github.com/vdemichev/diann-rpackage>) and DiannReport Generator Package was used to count the number of peptides quantified in each protein group (<https://github.com/kswovick/DIANN-Report-Generator>).⁹⁰ Representative heatmaps of normalized total spectral counts

were generated using Morpheus (<https://software.broadinstitute.org/morpheus>). Relative abundance was first normalized to the naïve control and were then normalized to the abundance of the sample's bait protein, resulting in a percentage of relative intensity as compared to the bait which was immunoprecipitated.

Immunofluorescence staining

Wildtype HCT116 cells or BRAT1 knockout lines were plated on coverslips in a 6-well plate. The next day, cells were gently washed with PBS and coverslips were moved to a 24-well plate. Cells were fixed by incubating in ice cold methanol for 5 minutes. After washing with PBS, samples were blocked for 30 minutes in 1% bovine serum albumin (Sigma-Aldrich, #A7906-50G) in PBS and 0.1% Tween 20 (Fisher Scientific, #BP337-500) at 37°C with humidity. Primary antibody probing was performed overnight at 1:100 in blocking buffer with rabbit anti-BRAT1 (Abclonal, #A12801). The next morning, samples were washed three times with PBS for 5 minutes. Probing of secondary antibody was performed for 1 hour in the dark with Texas Red-X goat anti-rabbit IgG (H+L) (Invitrogen, #T-6391). After another three five minute washes with PBS, coverslips were mounted on slides with Fluoroshield Mounting medium with DAPI (Abcam, #ab104139).

Image acquisition for BRAT1 expression and localization immunofluorescence

Mounted coverslips for BRAT1 and nuclei labeled HCT wild-type (WT) and BRAT1 knockout (KO) cells were imaged in a “grid” confocal regime using a structured illumination element (OptiGrid) inserted into an Olympus BX51 microscopy. The samples were illuminated with a Prior Lumen 200 Hg lamp source (Prior; #LM200B1-A) coupled into the microscope body with a liquid light coupler (Prior; #LM587). The broadband light source was filtered for 377 nm and 561 nm, respectively, with the following excitation filters (Semrock; #FF01-377/50-25, #FF01-561/14-25). The filtered light was focused onto the sample using either 20x (Olympus; cat: UPlanApo 0.70 NA air) or 60x (Olympus; cat: PlanApo 1.4NA oil) objectives. The resultant emissions from DAPI stained nuclei and Texas Red labeled BRAT1 were filtered through corresponding emission filters for DAPI (Semrock; cat: FF02-447/60-25) and Texas Red (FF01-609/54-25), respectively, and imaged onto a Hamatsu ORCA-ER detector. Images acquisition was controlled using Volocity Image Acquisition software (Quorum Technologies). The collected images were z-stacks acquired at either 1 μm or 0.5 μm axial steps for illumination with a 20x or 60x objective, respectively.

Image analysis for BRAT1 expression and localization immunofluorescence

Collected images were pre-processed on Volocity for noise filtering and flat field illumination correction. Then, the acquired z-stacks were analyzed for DAPI and Texas Red positive objects using a standard deviation dependent thresholding algorithm. DAPI-positive objects were segmented to identify individual nuclei, which was used as a proxy for number of cells in a field-of-view and normalization scheme for expression levels at a “per nuclei” or “per cell” basis. Texas Red positive objects were also segmented and determined to be nuclear or cytoplasmic based on spatial overlap of detected fluorescence with DAPI objects. BRAT1 localization was then determined as the fraction of cytoplasmic (non-overlapping with DAPI) BRAT1 volume, to nuclear (overlapping with DAPI) BRAT1 volume. BRAT1 expression levels was defined as the total intensity of BRAT1 objects normalized to the total BRAT1 volume and then normalized to a “per cell” quantal value based on the number of nuclei in a field-of-view. These values were then imported in GraphPad Prism for statistical analysis using a student's t test.

Protein expression and purification

Drosophila IntS11 and CG7044 were cloned into the same pFL vector and co-expressed in insect cells using Multibac technology (Geneva Biotech).⁹¹ A 6xHis tag was added to the N terminus of IntS11. Bacmids expressing IntS11 and CG7044 were generated in DH10EMBacY competent cells (Geneva Biotech) by transformation. Tni insect cells were grown in ESF 921 medium (Expression Systems) by shaking at 120 rpm at 27 °C until the density reached 2×10^6 cells $\cdot$ ml⁻¹. Cells were infected with 16 ml IntS11-CG7044 P2 virus and harvested after 48 h.

For purification, the cell pellet was resuspended and lysed by sonication in 100 ml of buffer containing 20 mM Tris HCl (pH 8.0), 250 mM NaCl, 2 mM β -mercaptoethanol (β ME), 5% (v/v) glycerol, and one tablet of protease inhibitor mixture (Sigma). The cell lysate was then centrifuged at 13,000 rpm for 40 min at 4 °C. The protein complex was purified from the supernatant via nickel affinity chromatography. The protein complex was further purified by a HiTrap Q column (GE Healthcare) and a Hiload 16/60 Superdex 200 column (GE Healthcare). The IntS11-CG7044 complex was concentrated to 1 mg $\cdot$ ml⁻¹ in a buffer containing 20 mM Tris HCl (pH 8.0), 100 mM NaCl, and 2 mM dithiothreitol (DTT), and stored at -80 °C. The protein concentration of the freshly purified samples was measured with a NanoDrop spectrophotometer (Thermo Fisher Scientific).

N-terminal 6xHis-tagged *Drosophila* IntS9 was cloned into the pSPL donor vector. *Drosophila* IntS11 and CG7044 were cloned into the pFL acceptor vector. These two vectors were fused together by Cre recombinase. One liter of Tni insect cells (2×10^6 cells $\cdot$ ml⁻¹) were infected with 15 ml of IntS9-IntS11-CG7044 P2 virus. For purification, the cell pellet was resuspended and lysed by sonication in 100 ml of buffer containing 20 mM Tris HCl (pH 8.0), 250 mM NaCl, 2 mM β ME, 5% (v/v) glycerol, and one tablet of protease inhibitor mixture (Sigma). The cell lysate was then centrifuged at 13,000 rpm for 40 min at 4 °C. The protein complex was purified from the supernatant via nickel affinity chromatography. The protein complex was further purified by a HiTrap Q column (GE Healthcare).

Human INTS9, INTS11 and BRAT1 were co-expressed in insect cells. N-terminal 6xHis-tagged BRAT1 and INTS11 were cloned into the same pFL vector. INTS9 was cloned into another pFL vector. One liter of Tni insect cells (2×10^6 cells $\cdot$ ml⁻¹) were co-infected

with 16 ml of BRAT1-INTS11 P2 virus and 16 ml of INTS9 P2 virus. The cells were harvested after 48 h. For purification, the cell pellet was resuspended and lysed by sonication in 100 ml of buffer containing 20 mM Tris HCl (pH 8.0), 250 mM NaCl, 2 mM β ME, 5% (v/v) glycerol, and one tablet of protease inhibitor mixture (Sigma). The cell lysate was then centrifuged at 13,000 rpm for 40 min at 4 °C. The protein complex was purified from the supernatant via nickel affinity chromatography. The protein complex was further purified by a Hiload 16/60 Superdex 200 column. The INTS9-INTS11-BRAT1 complex was concentrated to 1 mg·mL⁻¹ in a buffer containing 20 mM Tris HCl (pH 8.0), 200 mM NaCl, and 2 mM DTT, and stored at -80 °C.

The human INTS9 S283M and E280R/S283M/L287A/R290A mutations were introduced using site-directed mutagenesis PCR and verified by DNA sequencing. The mutant INTS9-INTS11-BRAT1 complexes were expressed and purified following the protocols for the wild-type complex.

K. lactis Ysh1 and Ipa1 were cloned into the 438B vector⁹² (Addgene, #55219). *K. lactis* Ydh1 was cloned into the 438A vector (Addgene, #55218), and Pta1 was cloned into the 438C vector (Addgene, #P55220). Ysh1, Ydh1, and Pta1 were combined into a single vector using the MacroBac technology.⁹² One liter of Tni insect cells (2×10^6 cells·mL⁻¹) were co-infected with 16 ml of Ysh1-Ydh1-Pta1 P2 virus and 16 ml of Ipa1 P2 virus. The cells were harvested after 48 h. For purification, the cell pellet was resuspended and lysed by sonication in 100 ml of buffer containing 20 mM Tris HCl (pH 8.0), 250 mM NaCl, 2 mM β ME, 5% (v/v) glycerol, and one tablet of protease inhibitor mixture (Sigma). The cell lysate was then centrifuged at 13,000 rpm for 40 min at 4 °C. The protein complex was purified from the supernatant via nickel affinity chromatography. The protein complex was further purified by a Hiload 16/60 Superdex 200 column. The Ysh1-Ipa1 complex was concentrated to 2 mg·mL⁻¹ in a buffer containing 20 mM Na HEPES (pH 7.4), 200 mM NaCl, and 2 mM DTT, and stored at -80 °C.

EM specimen preparation and data collection

Cryo-EM grids for the *Drosophila* IntS11-CG7044 complex were prepared by applying 3.5 μ L of protein sample at a concentration of 0.18 mg·mL⁻¹ to one side of a Quantifoil 400 mesh 1.2/1.3 gold grid with graphene oxide support film (Quantifoil). After 30 s, the grid was blotted for 1.5 s on the other side under 99% humidity and 20 °C using EM GP2 plunge freezer (Leica) and immediately plunged into liquid ethane. 1,603 image stacks were collected on a Titan Krios electron microscope at the New York Structural Biology Center, equipped with a K3 direct electron detector (Gatan) at 300 kV with a total dose of 51 e⁻ Å⁻² subdivided into 40 frames in 2 s exposure using Legikon.⁹³ The images were recorded at a nominal magnification of 81,000 $\times$ and a calibrated pixel size of 1.083 Å, with a defocus range from -1 to -2.5 μ m.

For the INTS9-INTS11-BRAT1 complex, a 3.5 μ L aliquot at 0.3 mg·mL⁻¹ was applied to a plasma-cleaned Quantifoil 300 mesh 1.2/1.3 grid (Quantifoil) with a locally applied gold coating (30 nm thickness setting, Safematic GmbH, Switzerland). After 5 s, the grid was blotted for 1.5 s and plunged into liquid ethane with a Vitrobot Mark VI (FEI) set at 20 °C and 100 % humidity. 4,030 image stacks were collected on a Titan Krios electron microscope at the Columbia University Cryogenic Electron Microscopy Center, equipped with a K3 direct electron detector (Gatan) at 300 kV with a total dose of 58.2 e⁻ Å⁻² subdivided into 50 frames in 2.5 s exposure using Legikon.⁹³ The images were recorded at a nominal magnification of 105,000 $\times$ and a calibrated pixel size of 0.83 Å, with a defocus range from -0.8 to -1.3 μ m.

Image processing

For both cryo-EM datasets, image stacks were motion-corrected and dose-weighted using RELION 3.1.⁹⁴ For the IntS11-CG7044 dataset, the CTF parameters were determined with CTFFIND4⁹⁵ in cryoSPARC.⁹⁶ 2,028,174 particles were auto-picked and subjected to 2D classification and *ab initio* reconstruction in cryoSPARC to generate eight initial 3D models. These models were then used in heterogeneous refinement against all the particles in cryoSPARC, and the good particles were then used in another round of heterogeneous refinement. 406,222 particles were then selected and imported to RELION for CTF refinement and Bayesian polishing. The polished particles were then imported back to cryoSPARC for homogeneous refinement, yielding a final map at 3.54 Å resolution.

For the INTS9-INTS11-BRAT1 dataset, the patch CTF parameters were determined with cryoSPARC. 1,805,449 particles were auto-picked and subjected to 2D classification in cryoSPARC. 323,946 particles in classes with recognizable features by visual inspection were used for *ab initio* reconstruction in cryoSPARC to generate two initial models. One of the initial models with 185,172 particles was selected and subjected to non-uniform refinement together with per-particle defocus refinement and global CTF refinement in cryoSPARC, yielding a final map at 3.21 Å resolution.

Model building

Atomic models for CG7044 and *Drosophila* IntS11 were built manually into the cryo-EM density with Coot.⁹⁷ A homology model for *Drosophila* IntS11 was generated with I-TASSER,⁹⁸ based on the structures of human CPSF73.⁹⁹ The starting models for human INTS9 and INTS11 were from the PDB entry 7BFP,³¹ and the starting model for human BRAT1 was from AlphaFold.⁶⁹ These models were fitted as a rigid body into the cryo-EM density map and manually revised using Coot. The atomic models were improved by real-space refinement with the program PHENIX.¹⁰⁰

AlphaFold prediction

AlphaFold-Multimer⁷² was used to predict models of protein complexes. The models were manually examined with PyMOL and Coot. The PAE plot was produced with PAE viewer.¹⁰¹ It should be noted that currently there are no structures in the Protein Data Bank that have a protein segment bound in the active site of CPSF73 or its homologs, and therefore the prediction of the UBE3D complex is unlikely to be biased by prior related structures.

CRISPR genomic editing

CRISPR/Cas9 was used to generate HCT116 BRAT1 knockout cell lines. Briefly, a 100 base pair genomic sequence flanking the BRAT1 start codon was input into CRISPOR to predict gRNAs (<http://crispor.tefor.net/crispor.py>).¹⁰² gRNA selection criteria was based upon specificity score based off number of off targets and minimal distance between the BRAT1 start codon and the Cas9 cut site. The selected gRNAs (Table S3) were cloned into a modified pSpCas9(BB)-2A-Puro (pX459), originally from Feng Zhang (Addgene, #62988), using an annealed oligonucleotide strategy. Our group modified the Flag tag in the vector to a myc tag using site directed mutagenesis prior to gRNA cloning. To transfect, 4×10^5 HCT116 cells were plated in a 24-well dish in complete media and 800 μ g plasmid with Lipofectamine 2000 (Invitrogen, #11668027) was added to the cells according to the manufacturer's instructions. After 24 hours, 1 μ g/mL puromycin was added to the media for 48 hours to select the cell population. Cell populations were expanded in normal growth medium and protein lysates collected by adding 2X SDS sample buffer (120mM Tris HCl pH6.8, 4% SDS, 200mM DTT, 20% Glycerol, and 0.02% Bromophenol blue) to cells in wells, incubated at room temperature with periodic swirling, boiled for 10 minutes at 95°C, and subjected to a short sonication. Western blotting was performed to verify decrease in BRAT1 expression across the heterogeneous cell population. Clonal lines were selected followed by additional western blotting and genotyping to confirm knockout. To genotype, genomic DNA was extracted from candidate clonal lines, and PCR performed to amplify the genomic region containing the BRAT1 start codon. PCR products were resolved on an agarose gel to verify disruption at the beginning of the gene on each allele. Finally, PCR products were cloned and sequenced to confirm identity.

RNA interference (*Drosophila* cells)

Double-stranded RNAs targeting the 5' and 3' UTRs of *Drosophila* IntS11 and the CDS of CG7044 were generated using *in vitro* transcription of PCR templates with the T7 promoter sequence on each end using MEGAscript kit (Thermo, #AMB13345). Primer sequences are in Table S3. For RNA interference experiments, 1.5×10^6 /ml of S2 cells were seeded into a 6-well plate with 10 μ g of dsRNA after being washed into serum free media. After a 1-hour incubation, 2 mL of complete growth medium was added. Samples were harvested for protein lysates 60 hours after the start of the incubation.

RT-qPCR

Total RNA was isolated using Trizol (Invitrogen, #15596026) following the manufacturer's instructions. cDNA was then reverse transcribed using M-MLV Reverse Transcriptase (Invitrogen, #28025013). Random hexamers were used for cDNA synthesis and RT-qPCR was then carried out in triplicate using Bio-Rad iTaq Universal SYBR Green Supermix (Bio-Rad, #1725120) and measured in AriaMx Real-Time PCR (qPCR) Instrument (Agilent, G8830A). All RT-qPCR primers are provided in Table S3.

Reporter cell line establishment

Reporter lines were generated as described previously.³³ Briefly, a reporter plasmid was generated by cutting pAAVS1-TLR targeting vector (Addgene, #64215) with *Cal1* and *PspXI* and replacing in the coding region of U7 small nuclear RNA, the 500 bp upstream of the transcription start site, and 50 bp downstream of the coding region followed by the GFP coding region and SV40 poly(A) signal. This was co-transfected with the pU6-(BbsI)CBh-Cas9-T2A-mCherry (Addgene, #64324) with the gRNA targeting the AAVS1 locus into HEK 293T cells. Lipofectamine 2000 was used to transfect in equal amounts of plasmid for 24 hours before a 2-day selection with 800 ng/ml puromycin. Cells were grown in growth medium without selection for a week before clonal selection.

Rescue experiments in Figure 5F were performed by transfecting in pSH-EFIREs-Puro Empty Vector, pSH-EFIREs-P-BRAT1-WT, pSH-EFIREs-P-BRAT1-AAA, pSH-EFIREs-P-BRAT1- Δ DCY, or pSH-EFIREs-P-BRAT1- Δ CTE. Plasmids were generated by replacing the *AtAFB2* coding sequence in pSH-EFIREs-P-*AtAFB2* (Addgene, #129715) with a multicloning site using oligonucleotides GGCCACGCGTTCTAGACAGCCAAACGGGTCAAACCTTG and GGCCACGCGTGAATTCAGCGCTAGCCTATAGTGAGT, and then cloning in the respective cDNAs. Mutations were introduced in oligonucleotide sequences (Table S3). Lysates were collected by adding 2X SDS loading buffer to cells in wells, incubating at room temperature with periodic swirling, boiling for 10 minutes at 95°C, and a brief sonication before loading into 10% SDS-PAGE gels for analysis.

Embryonic stem cell line establishment and organoid differentiation

Embryonic stem cell line H7 (WA07, WiCell) was used to generate syngeneic, BRAT1-knock out cell lines, by CRISPR-targeted disruption of BRAT1 gene using the gRNA and Cas9m plasmids described under *CRISPR genomic editing*. gRNAs (Table S3) were cloned into a modified pSpCas9(BB)-2A-Puro (pX459) vector (Addgene, #62988), and then transfected into H7 cells using Lipofectamine Stem transfection reagent (Invitrogen #STEM00003) according to manufacturer instructions. The next day stem cell cultures were rinsed with PBS and passaged as single cells using Accutase (Stem Cell Technologies #07922). To enable clonal selection, cells were plated on Matrigel (Corning #356254)-coated 100mm plates at 50 cells/cm² in Stemflex (Gibco, #A33494-01)

supplemented with CloneR (Stem Cell Technologies, #05888). One day later, medium was replaced with Stemflex supplemented with 1 μ g/ml puromycin to select for transfected cells. At the same time, colonies containing more than one cell were removed manually by pick-to-remove using an inverted microscope (Nikon TS-100) equipped with 4x Apochromat objective. After 4 days puromycin was removed from the medium. After an additional 4 days, surviving clones were manually harvested by pick-to-keep and expanded as separate clones in 24-well and then 6-well plates at which point cell pellets were harvested in parallel and analyzed by western blot for expression of BRAT1. Two clonal cell lines each were selected from control (no sgRNA) and Δ BRAT1 cultures for further analysis.

To examine early effects of BRAT1 loss on interneuron lineages in human ESC-derived cells, we adapted the established dual Smad inhibition protocol for the generation of neural organoids which were subsequently induced to acquire a ventral fate.^{61,103} Briefly, on day 0 ESCs were passaged into a single cell suspension using Accutase, and then plated at 9000 cells per well into ultra-low adhesion, v-bottom, 96 well cell culture plates (SBio #MS-9096VZ) in Stemflex medium supplemented with 10 μ M ROCK inhibitor Y-27632 (HelloBio #HB2297). Cells were allowed to settle and form aggregates. Neural induction is initiated on day 1 by addition of 10 μ M SB431542 (MedChemExpress #HY-10431) and 250nM LDN193189 (Tocris #4602) in the presence of Wnt-inhibitor XAV939 (2 μ M, Tocris #3748) in DMEM/F12 medium (Gibco #11330-057) supplemented with 15% knockout serum replacement (Gibco #10828010), Glutamax (Gibco #35050061), non-essential amino acids (Gibco #11140050), 100 μ M β -mercaptoethanol (Sigma #M3148). Going forward medium was exchanged every 2 days. On days 5-8 organoids were switched to a 50:50 mix of DMEM/F12 and Neurobasal (Gibco #21103049) medium supplemented with N2 (Gibco #17520248), Glutamax, penicillin-streptomycin antibiotic (Gibco #15140122), 10 μ M SB431542, 250nM LDN193189 and 2 μ M XAV939. On day 9 SB431542, LDN193189 and XAV939 are omitted and a ventral fate is induced by addition of the Sonic Hedgehog agonist SAG (0.5 μ M, Cayman Chemicals #11915) and 1 μ M pumorphamine (Tocris, #45-511-0). Ventral organoids were then harvested on day 14.

Stem cell staining

Cell colonies of control and Δ BRAT1 embryonic stem cells were grown in low-profile, 4-well dishes (Nunc, #179830) in stem cell conditions prior to harvesting for immunofluorescent analysis. For live staining with Tra-1-81 (1:200, mouse IgM, EMD Millipore #MAB4381), Tra-1-60 (1:200, mouse IgM, Invitrogen #MA1-023) and SSEA4 (1:100, mouse IgG3, Abcam #ab16287), antibodies were added directly to the cell culture medium and incubated for 30 minutes prior to washing with PBS and fixation in 4% paraformaldehyde (PFA)/PBS for 10 min at room temperature. For labeling with OCT-4, SOX2 and NANOG, cultures were fixed in 4% PFA/PBS for 10 minutes at room temperature, and rinsed in Wash Medium (PBS, 5% (v/v) goat serum), then permeabilized for 10 minutes in 0.5% (v/v) Triton X-100 in Wash Medium, and blocked for one hour in Staining Buffer (1% (v/v) bovine serum albumin, 5% (v/v) goat serum and 0.2% (v/v) Triton X-100). OCT-4 (1:200, rabbit host, Cell Signaling Technologies #2750), SOX2 (1:500, rabbit monoclonal, Cell Signaling Technologies #3579) and NANOG (1:200, rabbit monoclonal, Cell Signaling Technologies #4903) antibody labeling was performed overnight at 4°C in Staining Buffer. Following primary antibody staining, cells were washed with Wash Medium and incubated with appropriate secondary antibodies diluted in Staining Buffer for 1 hour: Goat anti-rabbit IgG Alexa Fluor 488 (Invitrogen, #A-11034), goat anti-rabbit IgG Alexa Fluor 568 (Invitrogen #A-11036), goat anti-mouse IgM Alexa Fluor 488 (Invitrogen #A-21042), goat anti-mouse IgM Alexa Fluor 568 (Invitrogen #A-21043), goat anti-mouse IgG(H+L) Alexa Fluor 568 (Invitrogen #A11031). Samples were then washed three times in Wash Medium and then stored in PBS with 0.05% sodium azide prior to fluorescent imaging on a Evos FL AMF4300 (Invitrogen #12-563-460) inverted epi-fluorescent microscope equipped for imaging ex470/em525nm (GFP) and ex531/em593nm (RFP) fluorophores.

Small-scale extract and co-immunoprecipitation

Small scale nuclear extract and co-immunoprecipitations in Figure 7F were performed similarly to described above. HEK 293T cells were transfected with equal amounts of pcDNA-HA-CPSF73 and pcDNA-FLAG empty vector, pcDNA-FLAG-UBE3D-WT, pcDNA-FLAG-UBE3D- Δ loop-1, pcDNA-FLAG-UBE3D- Δ loop-2, or pcDNA-FLAG-UBE3D-AAAA using Lipofectamine 2000 according to the manufacturer's directions (Invitrogen, #11668027). Cells were harvested from a 6-well plate by washing with 1x PBS, then lysing with low salt lysis buffer (150mM NaCl, 1% NP-40, 100mM Tris-HCl pH8.8) and incubating on ice for 30 minutes. Samples were centrifuged at 20,000g for 5 minutes at 4°C. The supernatant was removed for co-immunoprecipitation and a small aliquot saved for input blots. Similarly to above, extract was incubated with anti-Flag M2 affinity agarose slurry (Sigma, #A2220) equilibrated in IP wash buffer (20mM Na HEPES pH7.4, 100mM KCl, 10% Glycerol, 0.1% NP-40) and rotated for 4 hours at 4°C. After a 10 minute incubation on ice, samples were centrifuged at 500g for 3 minutes at 4°C. Four washes were performed by removing supernatant, adding 1 mL IP wash buffer, rotating for 5 minutes at 4°C, and spin at 500g for 3 minutes at 4°C. After the four washes, the supernatant was removed, and 2x SDS loading buffer was added the beads before a ten-minute boil at 95°C, and loading onto 10% SDS-PAGE gels for analysis.

QUANTIFICATION AND STATISTICAL ANALYSIS

RT-qPCR quantification and analysis

Data were analyzed using the $\Delta\Delta$ Ct method with 7sk as the reference gene and wildtype HCT116 or Day 0 H7 ESC cells as the control, as described previously.⁵⁷ Results are shown from biologically independent replicates, depicting averages and standard deviations (mean $\pm$ SD, N=3).

RNA-seq library generation and mapping

Embryonic stem cells and neural organoids were washed once in PBS prior to harvest, and up to 2×10^6 cells or 20 organoids were lysed in 500ul RNA lysis buffer (Zymo Research #R1060) using 2 mm Bashing Beads (Zymo Research #S6003) in a bead mill homogenizer (VWR #19-2141T) with a 10 second pulse. RNA was extracted using affinity column purification and on column DNase treatment (Zymo Research, #R1054). RNA quality and concentration was confirmed with a NanoDrop One^C (ThermoFisher, #ND-ONE-W). Input RNA was subjected to polyA selection using NEBNext Poly(A) mRNA Magnetic Isolation Module (NEB, #E7490S). 2 μ g of RNA from each sample was used for the kit purification and were eluted in 10 mM Tris HCl pH7.5 before proceeding to library generation. RNA-seq libraries were generated (3 independent biological replicates per condition) using the Click-seq library preparation method with a 1:35 azido-nucleotide ratio.¹⁰⁴ Libraries were sequenced using a single-end 150 bp cycle run on an Illumina NovaSeq 6000.

Raw reads generated from the Illumina basecalls were demultiplexed using bcl2fastq version 2.20.0. Quality filtering and adapter removal are performed using FastP version 0.23.1⁷⁹ with the following parameters: “--length_required 35 --cut_front_window_size 1 --cut_front_mean_quality 13 --cut_front --cut_tail_window_size 1 --cut_tail_mean_quality 13 --cut_tail -y -r”. Processed/cleaned reads were then mapped to the GRCh38/gencode38 (Mouse OR Human) reference¹⁰⁵ using STAR_2.7.9a⁸⁰ with the following parameters: “--twopass Mode Basic --runMode alignReads --outSAMtype BAM Unsorted --outSAMstrandField intronMotif --outFilterIntronMotifs RemoveNoncanonical --outReadsUnmapped Fastx.” Gene level read quantification was derived using the subread-2.0.1 package (featureCounts)⁸¹ with a GTF annotation file (GRCh38/gencode42) and the following parameters for stranded RNA libraries “-s 2 -t exon -g gene_name.” Differential expression analysis was performed using DESeq2-1.34.0 with a P-value threshold of 0.05 within R version 4.0.2 (<https://www.R-project.org/>).⁸² PCAplot was created within R using the pcaExplorer to measure sample expression variance.⁸³ Gene ontology analyses were performed using the EnrichR package^{84,106,107} and ShinyGO 0.77.⁸⁷ Volcano plots and dot plots were created using ggplot2.⁸⁵ Gene tracks were generated using Integrated Genome Viewer.⁸⁶

Genomic statistical tests

For RNA-seq, statistical significance was determined using Mann-Whitney pairwise tests unless otherwise noted. The details of violin plots, statistical tests and error bars are explained in their respective figure legends.